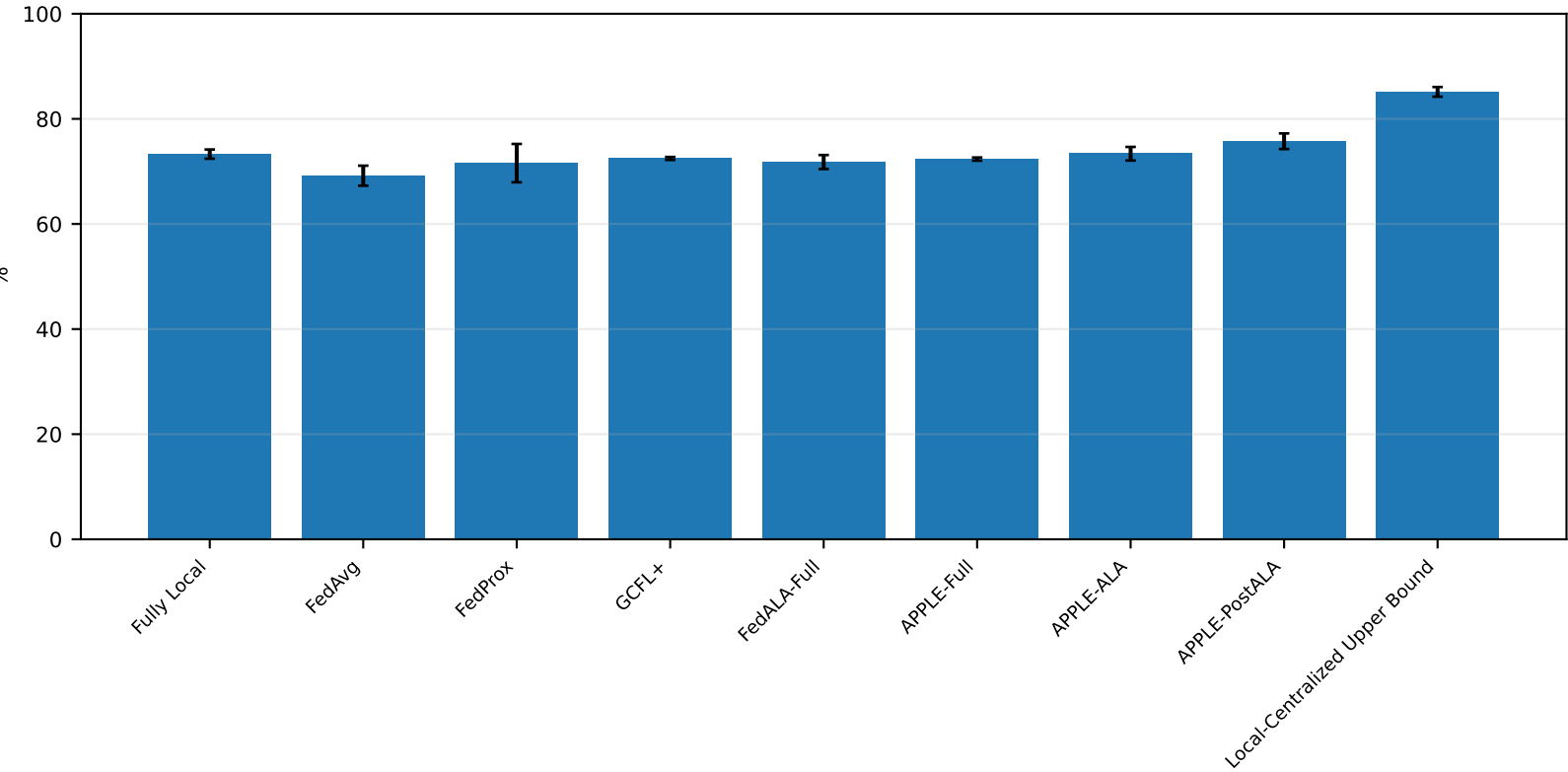


Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

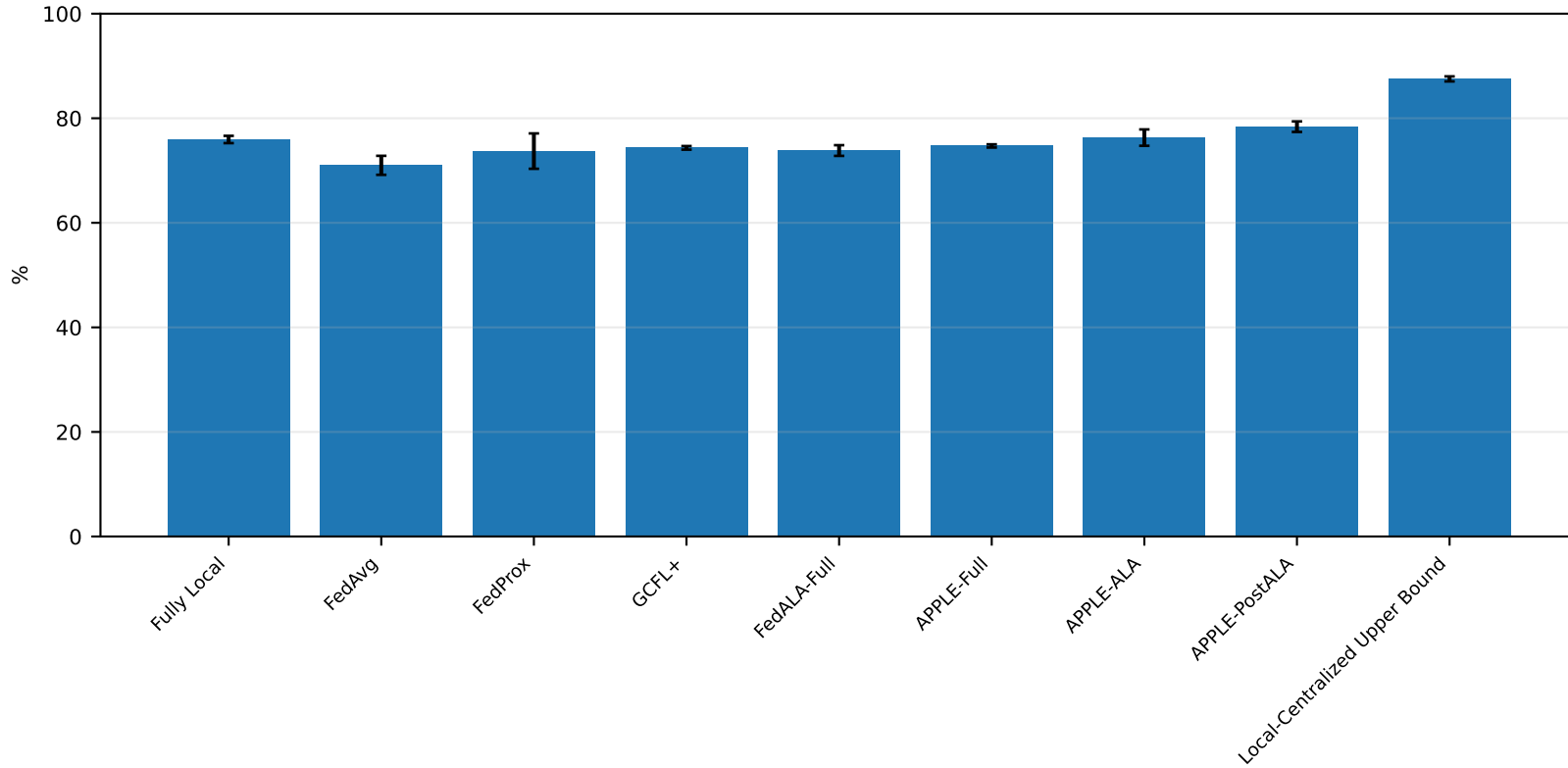
Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.3% ± 0.9%	75.9% ± 0.7%	81.4% ± 0.8%	79.2% ± 1.1%	90.7% ± 1.3%	86.5% ± 0.1%	76.0% ± 1.3%	73.0% ± 2.9%	53.5% ± 3.0%	—	3
FedAvg	69.2% ± 1.9%	71.0% ± 1.8%	83.0% ± 0.5%	81.9% ± 0.7%	82.2% ± 3.0%	76.0% ± 3.6%	70.0% ± 2.2%	68.2% ± 4.3%	58.6% ± 1.6%	71.7 ± 4.7	3
FedProx	71.6% ± 3.6%	73.7% ± 3.4%	85.8% ± 1.2%	84.7% ± 1.7%	85.0% ± 4.8%	81.9% ± 5.2%	72.0% ± 1.0%	70.0% ± 2.9%	59.7% ± 5.5%	78.0 ± 0.0	3
GCFL+	72.5% ± 0.3%	74.3% ± 0.3%	86.5% ± 3.9%	85.4% ± 3.5%	81.9% ± 5.4%	83.6% ± 1.6%	75.1% ± 0.5%	71.0% ± 2.3%	60.2% ± 0.5%	72.7 ± 5.1	3
FedALA-Full	71.8% ± 1.3%	73.8% ± 1.0%	82.6% ± 1.8%	81.6% ± 2.7%	86.1% ± 2.6%	80.1% ± 1.6%	72.7% ± 2.4%	71.3% ± 3.5%	58.8% ± 1.6%	68.3 ± 5.0	3
APPLE-Full	72.3% ± 0.3%	74.7% ± 0.3%	80.4% ± 2.3%	80.0% ± 2.4%	87.7% ± 2.1%	85.0% ± 2.4%	68.5% ± 2.9%	72.3% ± 0.4%	60.0% ± 1.9%	74.7 ± 2.1	3
APPLE-ALA	73.4% ± 1.3%	76.3% ± 1.6%	84.8% ± 3.5%	82.8% ± 3.0%	92.1% ± 3.3%	84.1% ± 1.3%	76.7% ± 3.7%	71.6% ± 1.4%	57.0% ± 1.0%	63.0 ± 10.4	3
APPLE-PostALA	75.7% ± 1.5%	78.4% ± 1.0%	87.0% ± 0.8%	86.1% ± 1.9%	89.0% ± 0.9%	91.0% ± 1.8%	77.6% ± 1.6%	73.4% ± 2.2%	60.9% ± 4.1%	76.7 ± 3.2	3
Local-Centralized Upper Bound	85.1% ± 0.9%	87.5% ± 0.5%	95.2% ± 1.5%	93.7% ± 0.8%	98.2% ± 0.1%	94.9% ± 4.4%	91.3% ± 2.9%	83.5% ± 2.7%	69.8% ± 5.7%	—	3

Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test

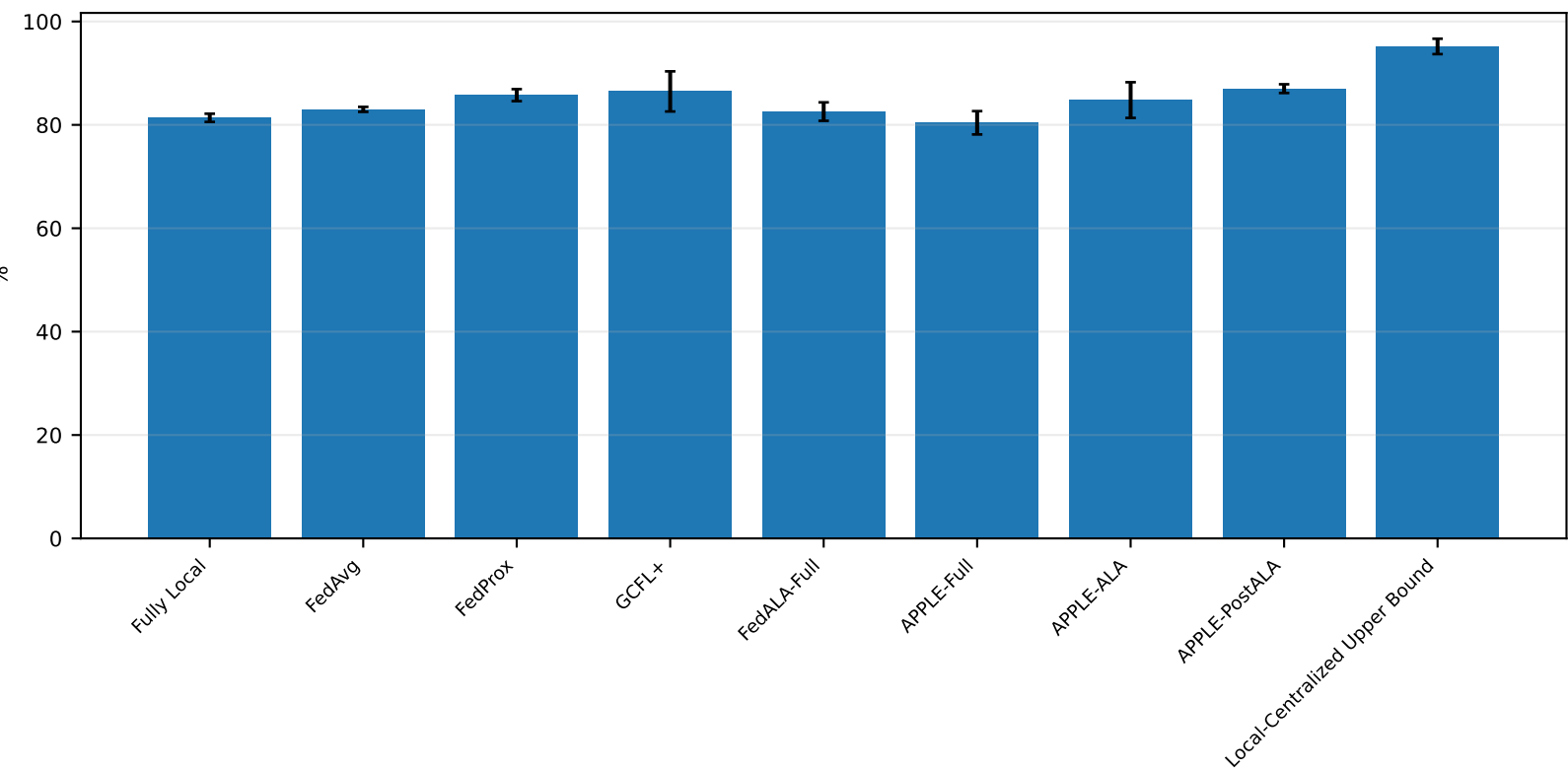
Micro-F1



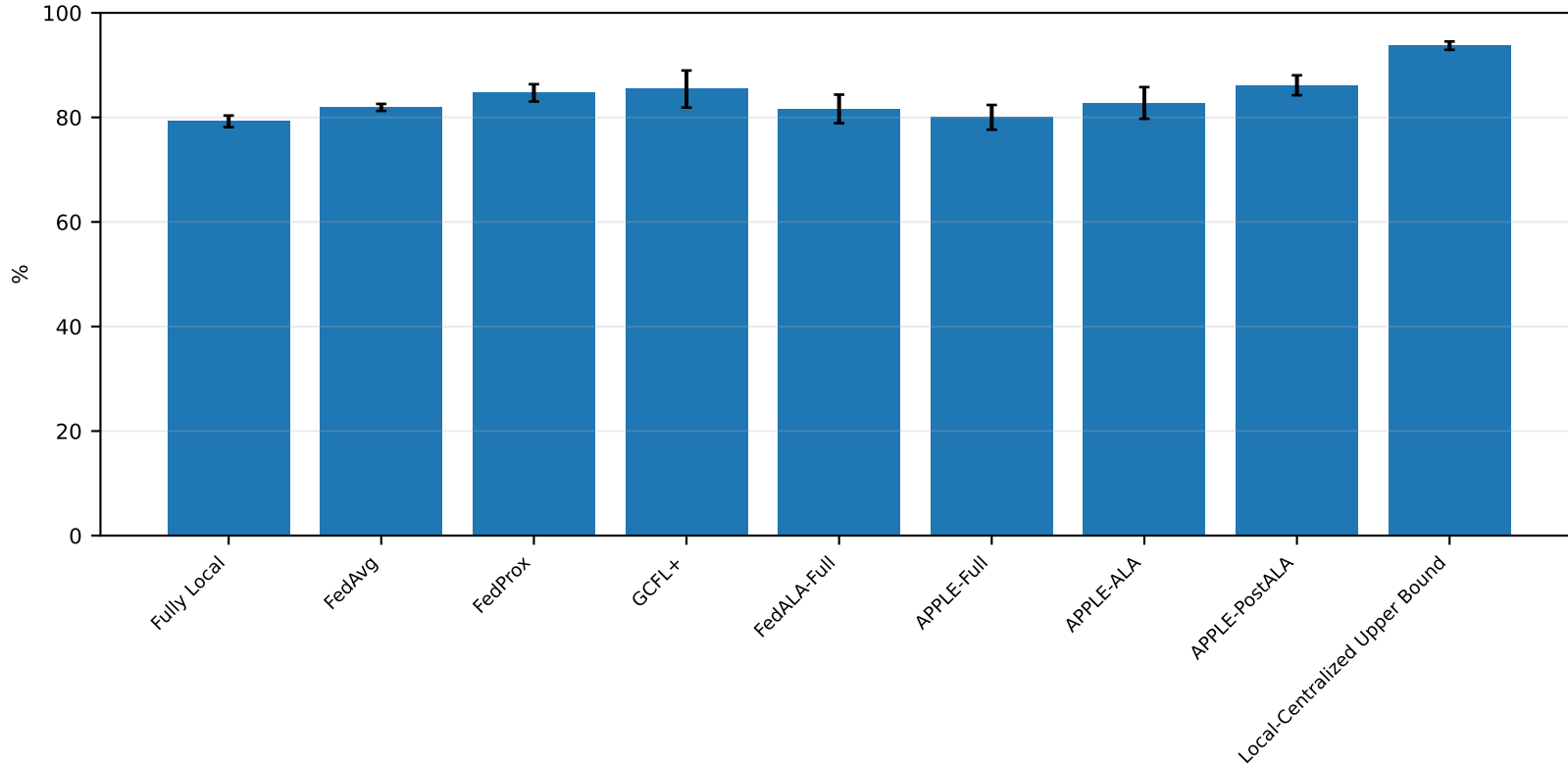
Macro Pos-F1



Micro PR-AUC



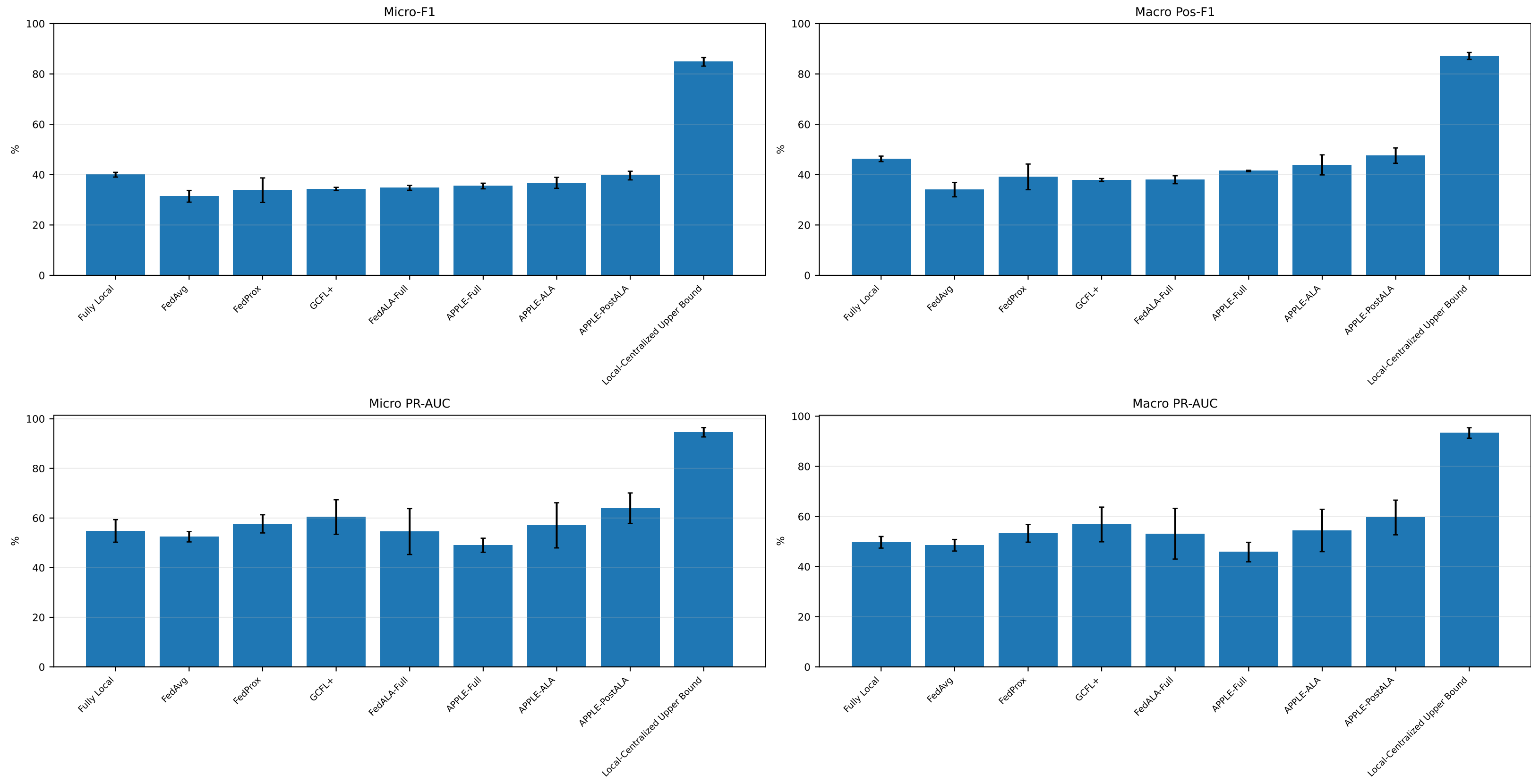
Macro PR-AUC



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	40.0% ± 0.9%	46.3% ± 1.1%	54.8% ± 4.5%	49.7% ± 2.3%	66.7% ± 4.1%	61.5% ± 4.0%	41.3% ± 3.4%	38.6% ± 3.0%	23.3% ± 1.2%	—	3
FedAvg	31.4% ± 2.3%	34.0% ± 2.8%	52.4% ± 2.1%	48.5% ± 2.3%	42.8% ± 8.4%	40.8% ± 1.6%	33.5% ± 2.5%	30.7% ± 4.6%	22.4% ± 0.9%	68.3 ± 9.5	3
FedProx	33.8% ± 4.9%	39.1% ± 5.1%	57.6% ± 3.7%	53.3% ± 3.5%	58.3% ± 3.3%	48.8% ± 13.0%	33.8% ± 1.7%	31.8% ± 4.0%	22.9% ± 4.4%	69.0 ± 9.5	3
GCFL+	34.3% ± 0.6%	37.9% ± 0.6%	60.4% ± 7.0%	56.8% ± 6.9%	45.5% ± 4.5%	50.0% ± 4.9%	37.7% ± 0.9%	32.9% ± 2.1%	23.3% ± 0.3%	70.0 ± 6.1	3
FedALA-Full	34.8% ± 1.0%	38.0% ± 1.6%	54.5% ± 9.3%	53.1% ± 10.1%	51.4% ± 8.8%	45.8% ± 1.9%	37.5% ± 5.4%	32.7% ± 2.4%	22.4% ± 1.1%	68.7 ± 1.5	3
APPLE-Full	35.5% ± 1.1%	41.5% ± 0.3%	49.0% ± 2.8%	45.8% ± 3.9%	64.2% ± 1.9%	53.2% ± 5.3%	31.6% ± 2.3%	34.6% ± 1.6%	23.7% ± 1.7%	73.7 ± 4.5	3
APPLE-ALA	36.8% ± 2.2%	43.9% ± 4.0%	57.0% ± 9.1%	54.4% ± 8.4%	71.2% ± 13.8%	51.2% ± 2.3%	40.6% ± 6.0%	34.3% ± 1.8%	22.0% ± 0.5%	63.0 ± 10.4	3
APPLE-PostALA	39.6% ± 1.7%	47.6% ± 3.0%	64.0% ± 6.1%	59.6% ± 6.9%	65.4% ± 6.4%	68.0% ± 6.8%	44.2% ± 7.1%	35.8% ± 2.6%	24.4% ± 3.2%	76.7 ± 3.2	3
Local-Centralized Upper Bound	84.8% ± 1.7%	87.1% ± 1.4%	94.6% ± 1.9%	93.3% ± 2.1%	96.4% ± 3.1%	95.9% ± 2.2%	91.4% ± 2.2%	82.8% ± 2.4%	69.3% ± 6.0%	—	3

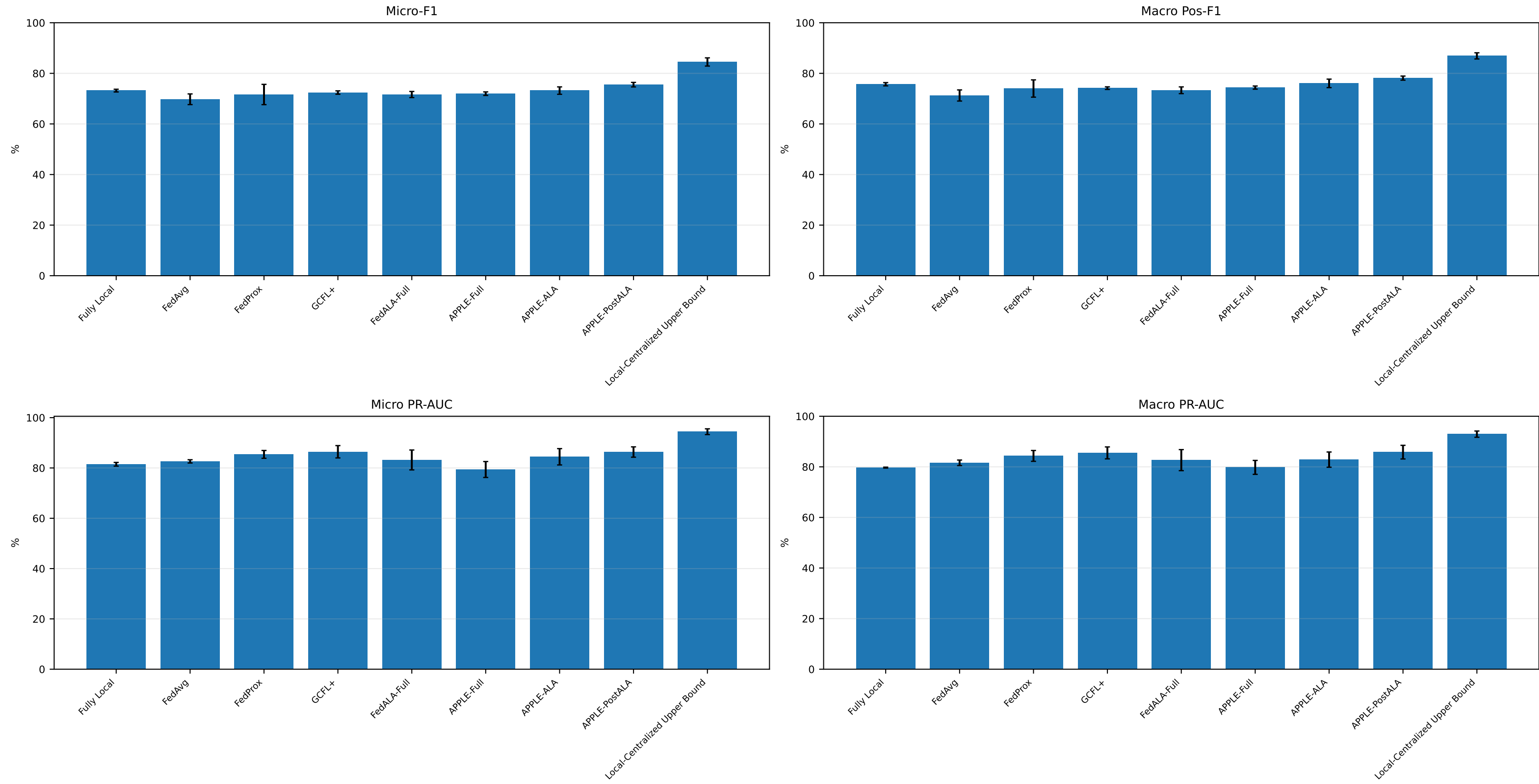
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.2% ± 0.5%	75.7% ± 0.6%	81.5% ± 0.7%	79.7% ± 0.2%	89.7% ± 0.6%	87.0% ± 0.8%	75.5% ± 3.4%	73.9% ± 1.3%	52.4% ± 2.0%	—	3
FedAvg	69.8% ± 2.1%	71.3% ± 2.2%	82.6% ± 0.6%	81.6% ± 1.1%	78.3% ± 4.9%	78.2% ± 1.6%	71.4% ± 1.6%	69.3% ± 4.8%	59.1% ± 1.5%	68.3 ± 9.5	3
FedProx	71.6% ± 4.0%	74.0% ± 3.4%	85.4% ± 1.6%	84.3% ± 2.1%	87.4% ± 0.5%	81.5% ± 6.8%	71.4% ± 1.1%	70.2% ± 3.9%	59.6% ± 6.1%	69.0 ± 9.5	3
GCFL+	72.4% ± 0.7%	74.2% ± 0.5%	86.4% ± 2.5%	85.5% ± 2.4%	80.6% ± 2.7%	83.4% ± 2.5%	75.1% ± 0.9%	71.6% ± 2.8%	60.2% ± 0.7%	70.0 ± 6.1	3
FedALA-Full	71.6% ± 1.2%	73.3% ± 1.3%	83.2% ± 4.0%	82.6% ± 4.2%	83.9% ± 4.4%	80.1% ± 1.7%	73.7% ± 4.0%	70.4% ± 2.5%	58.5% ± 2.1%	68.7 ± 1.5	3
APPLE-Full	72.0% ± 0.7%	74.4% ± 0.6%	79.4% ± 3.2%	79.8% ± 2.8%	87.9% ± 1.5%	84.2% ± 2.4%	68.6% ± 2.5%	71.4% ± 0.2%	59.7% ± 1.7%	73.7 ± 4.5	3
APPLE-ALA	73.2% ± 1.5%	76.0% ± 1.7%	84.4% ± 3.2%	82.9% ± 3.0%	91.8% ± 4.0%	83.7% ± 1.2%	76.1% ± 4.2%	71.3% ± 1.4%	57.2% ± 1.1%	63.0 ± 10.4	3
APPLE-PostALA	75.5% ± 0.9%	78.1% ± 0.8%	86.3% ± 2.1%	85.8% ± 2.7%	88.9% ± 1.8%	90.0% ± 2.8%	77.3% ± 2.2%	73.4% ± 1.8%	61.0% ± 3.7%	76.7 ± 3.2	3
Local-Centralized Upper Bound	84.5% ± 1.6%	86.9% ± 1.2%	94.4% ± 1.1%	92.9% ± 1.2%	95.4% ± 3.5%	97.0% ± 1.8%	90.9% ± 1.7%	82.8% ± 1.7%	68.4% ± 5.5%	—	3

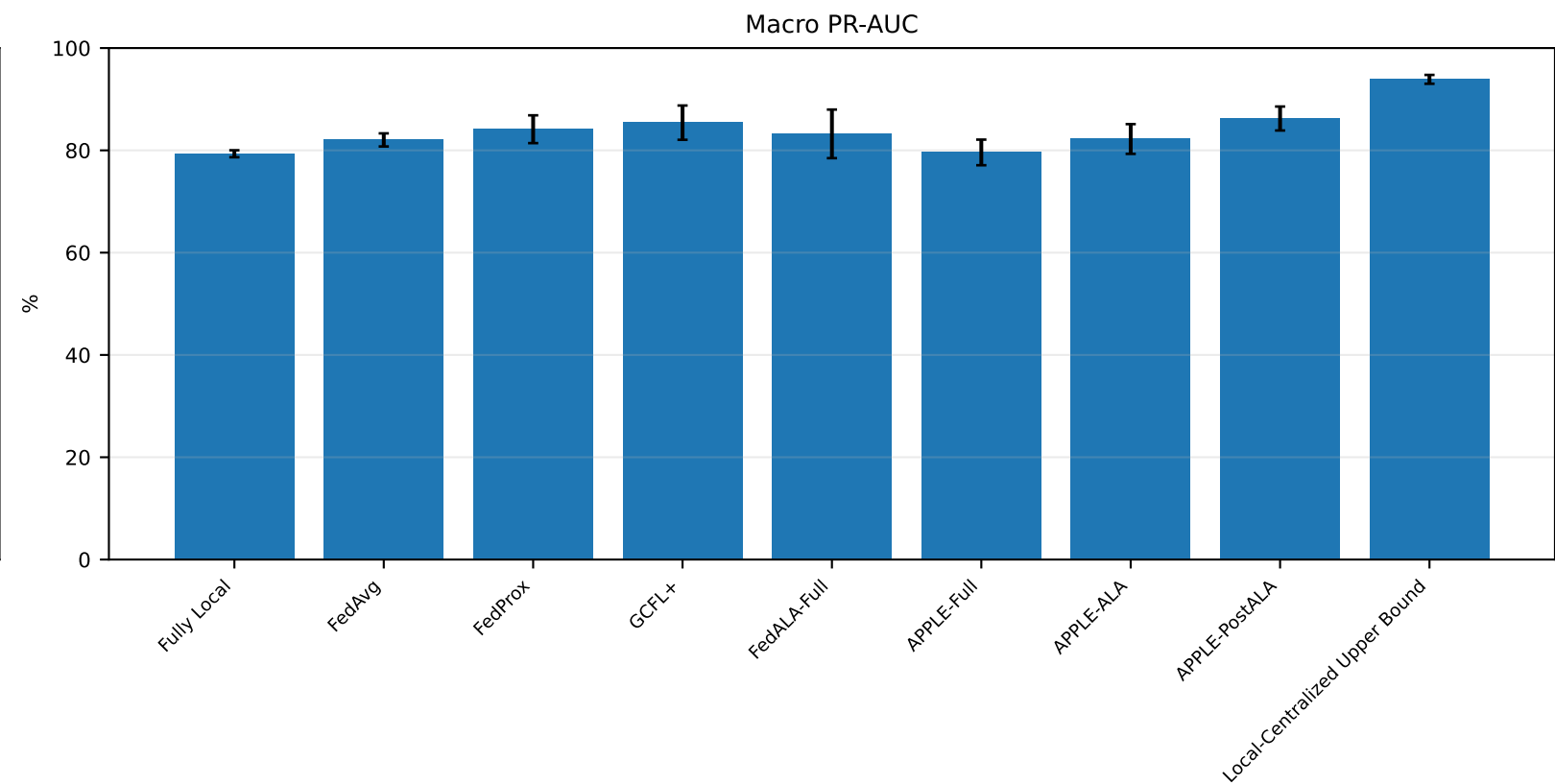
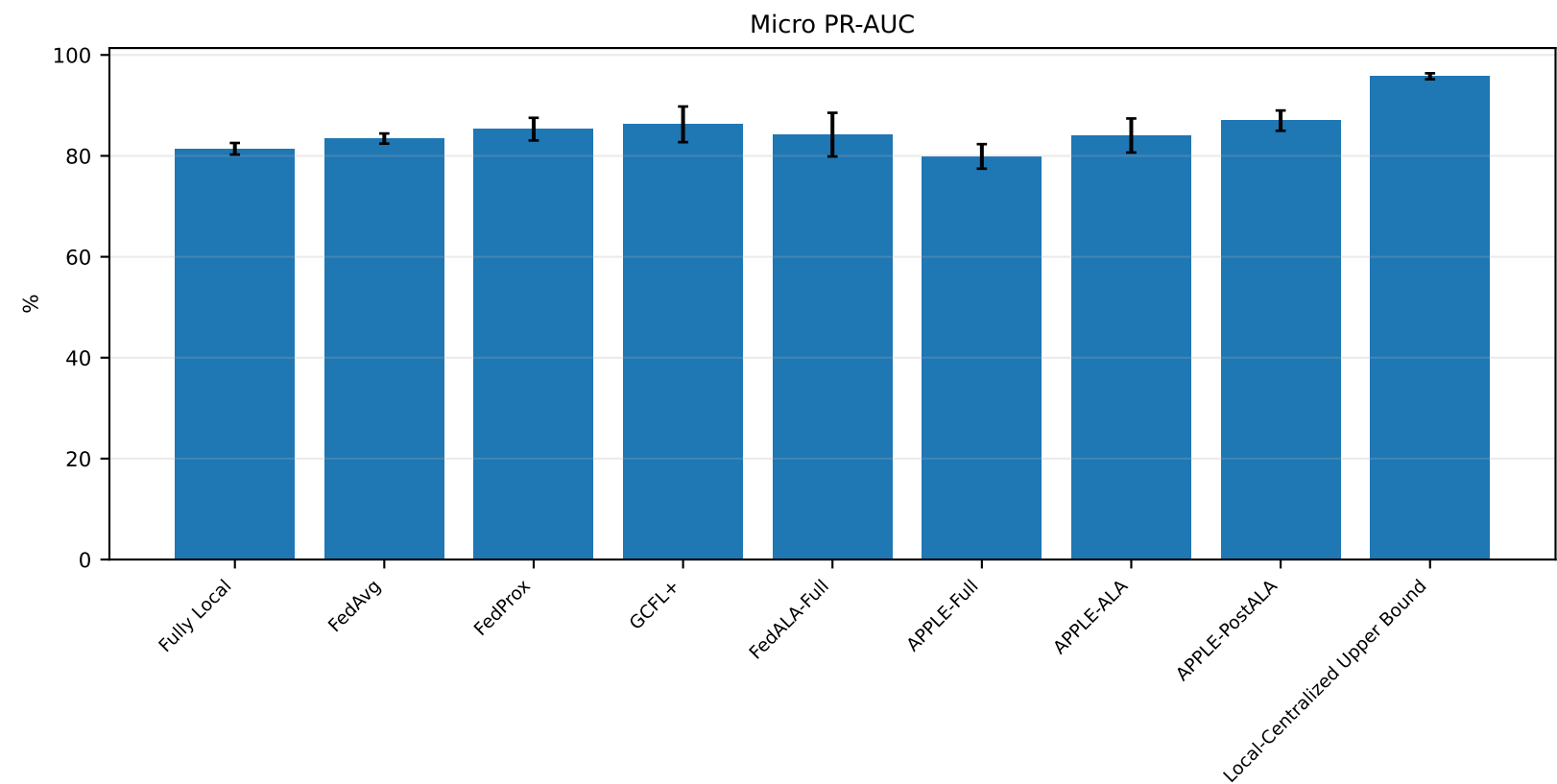
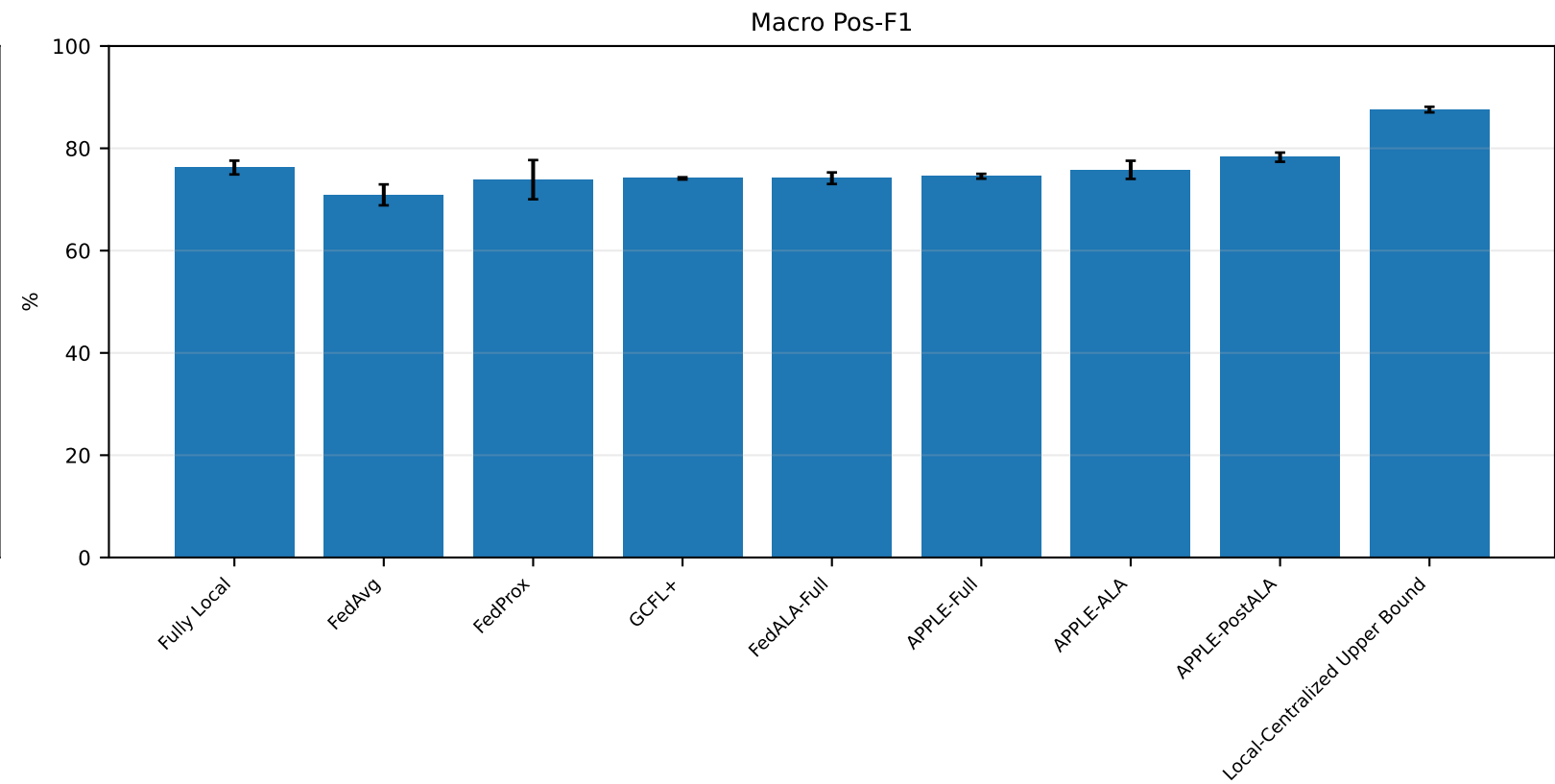
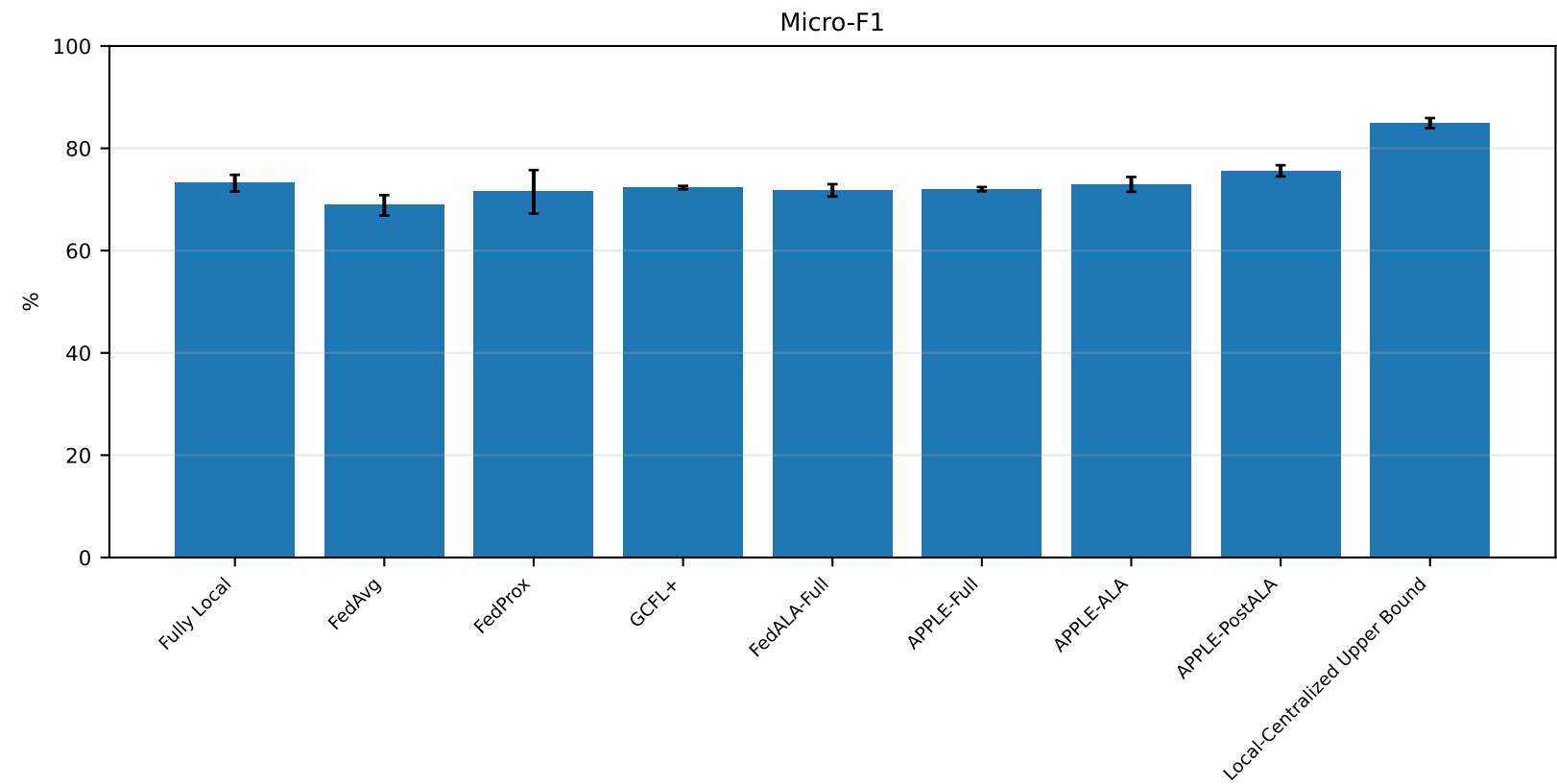
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pos_f1 | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	73.2% ± 1.6%	76.2% ± 1.3%	81.4% ± 1.1%	79.3% ± 0.7%	91.1% ± 1.5%	87.9% ± 1.6%	75.9% ± 0.8%	73.6% ± 4.3%	52.6% ± 3.0%	—	3
FedAvg	68.9% ± 2.0%	70.9% ± 2.0%	83.4% ± 1.0%	82.0% ± 1.3%	83.2% ± 3.6%	76.4% ± 4.0%	70.1% ± 2.5%	67.1% ± 4.1%	57.7% ± 2.4%	67.7 ± 10.7	3
FedProx	71.5% ± 4.3%	73.9% ± 3.8%	85.3% ± 2.2%	84.1% ± 2.7%	87.8% ± 2.4%	81.3% ± 6.5%	71.4% ± 1.9%	69.3% ± 3.9%	59.6% ± 5.8%	71.7 ± 11.0	3
GCFL+	72.3% ± 0.3%	74.1% ± 0.2%	86.2% ± 3.5%	85.4% ± 3.4%	81.4% ± 4.7%	83.4% ± 1.2%	74.8% ± 0.5%	70.8% ± 2.8%	60.2% ± 0.4%	72.7 ± 5.1	3
FedALA-Full	71.8% ± 1.2%	74.2% ± 1.1%	84.2% ± 4.3%	83.2% ± 4.7%	88.0% ± 0.6%	82.0% ± 3.4%	72.7% ± 1.5%	69.5% ± 3.2%	58.6% ± 1.7%	70.7 ± 2.1	3
APPLE-Full	72.0% ± 0.4%	74.5% ± 0.5%	79.9% ± 2.4%	79.6% ± 2.5%	89.3% ± 1.3%	84.2% ± 2.4%	67.9% ± 2.7%	71.9% ± 1.2%	59.4% ± 1.0%	75.0 ± 2.6	3
APPLE-ALA	73.0% ± 1.4%	75.8% ± 1.8%	84.0% ± 3.4%	82.2% ± 2.9%	91.1% ± 4.6%	83.3% ± 2.0%	76.4% ± 3.2%	71.3% ± 1.5%	56.8% ± 1.1%	63.0 ± 10.4	3
APPLE-PostALA	75.6% ± 1.1%	78.3% ± 0.9%	87.0% ± 2.0%	86.2% ± 2.3%	88.9% ± 1.8%	91.1% ± 3.2%	77.2% ± 2.0%	73.3% ± 2.8%	60.8% ± 4.0%	76.7 ± 3.2	3
Local-Centralized Upper Bound	84.9% ± 1.0%	87.6% ± 0.5%	95.8% ± 0.6%	93.9% ± 0.9%	98.4% ± 0.5%	97.3% ± 1.8%	90.3% ± 2.7%	83.9% ± 2.6%	68.0% ± 2.8%	—	3

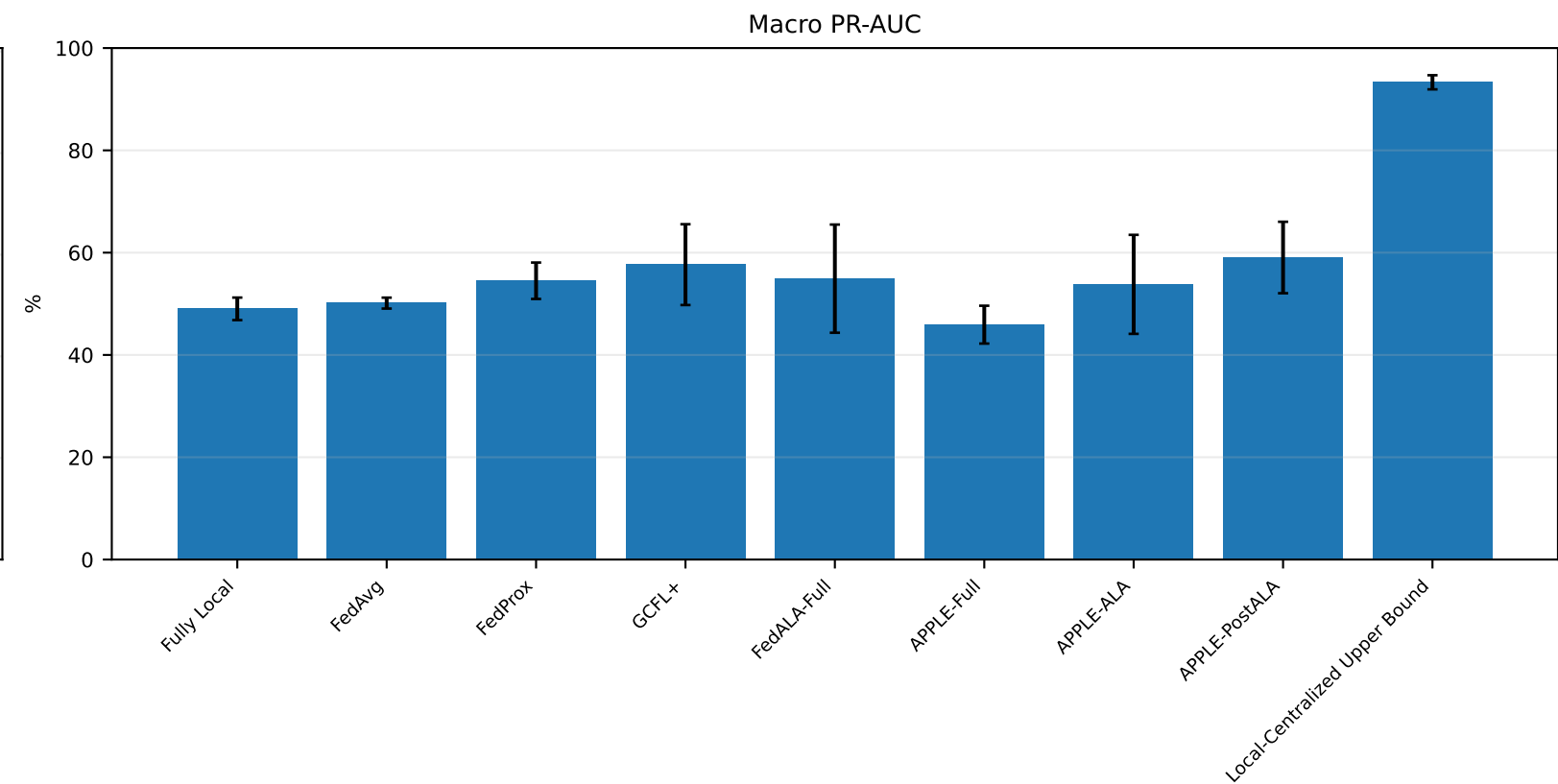
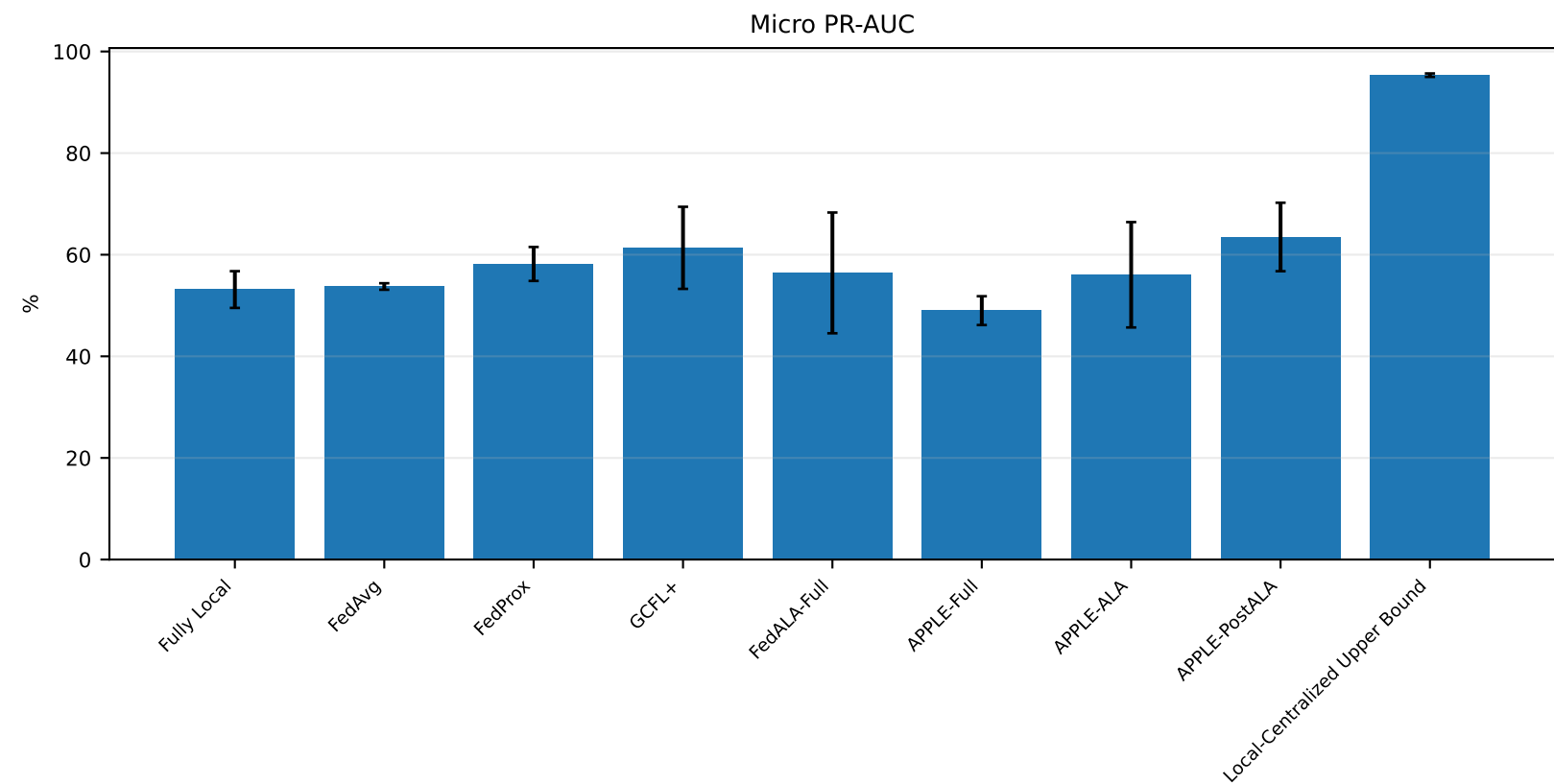
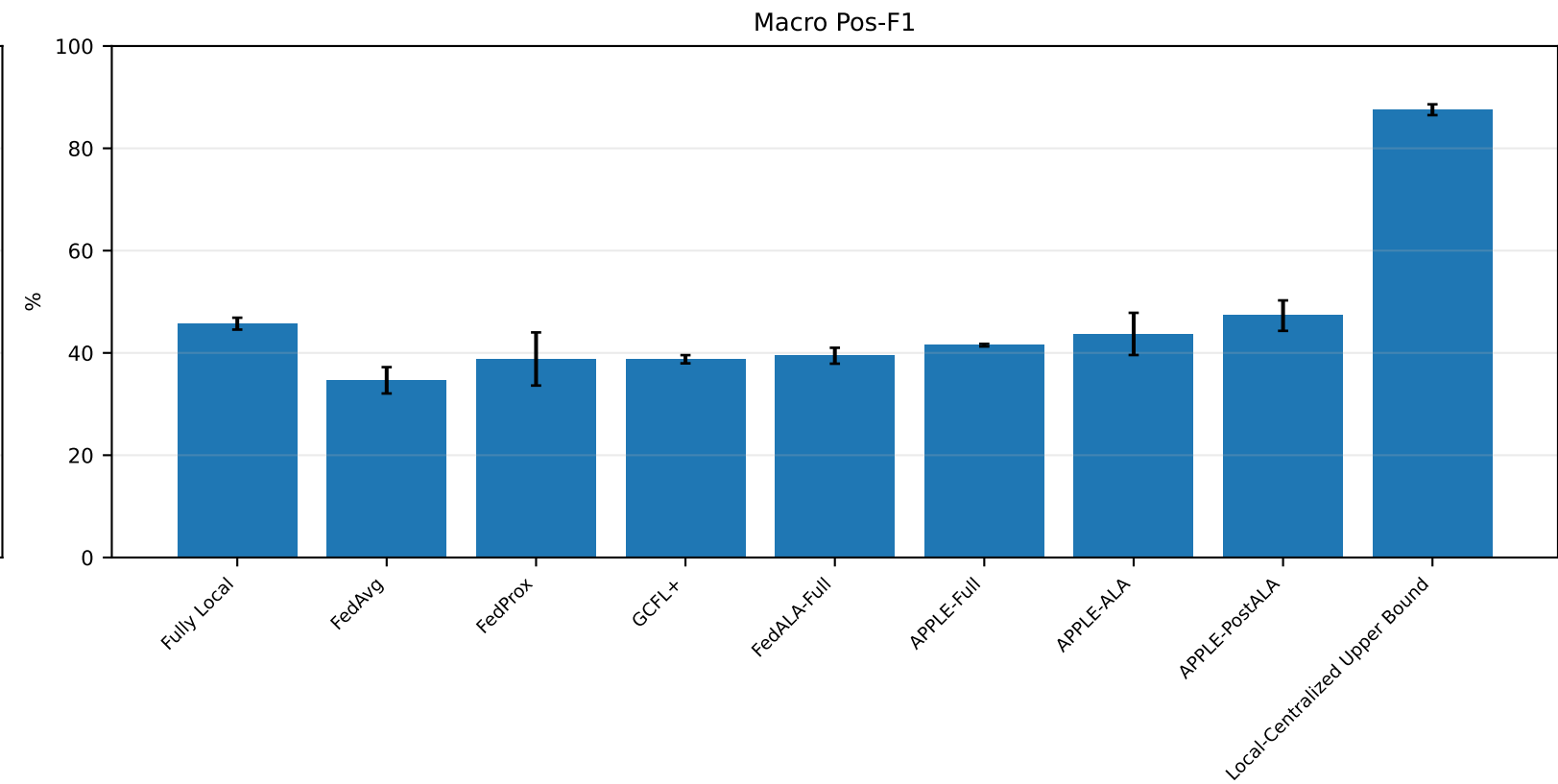
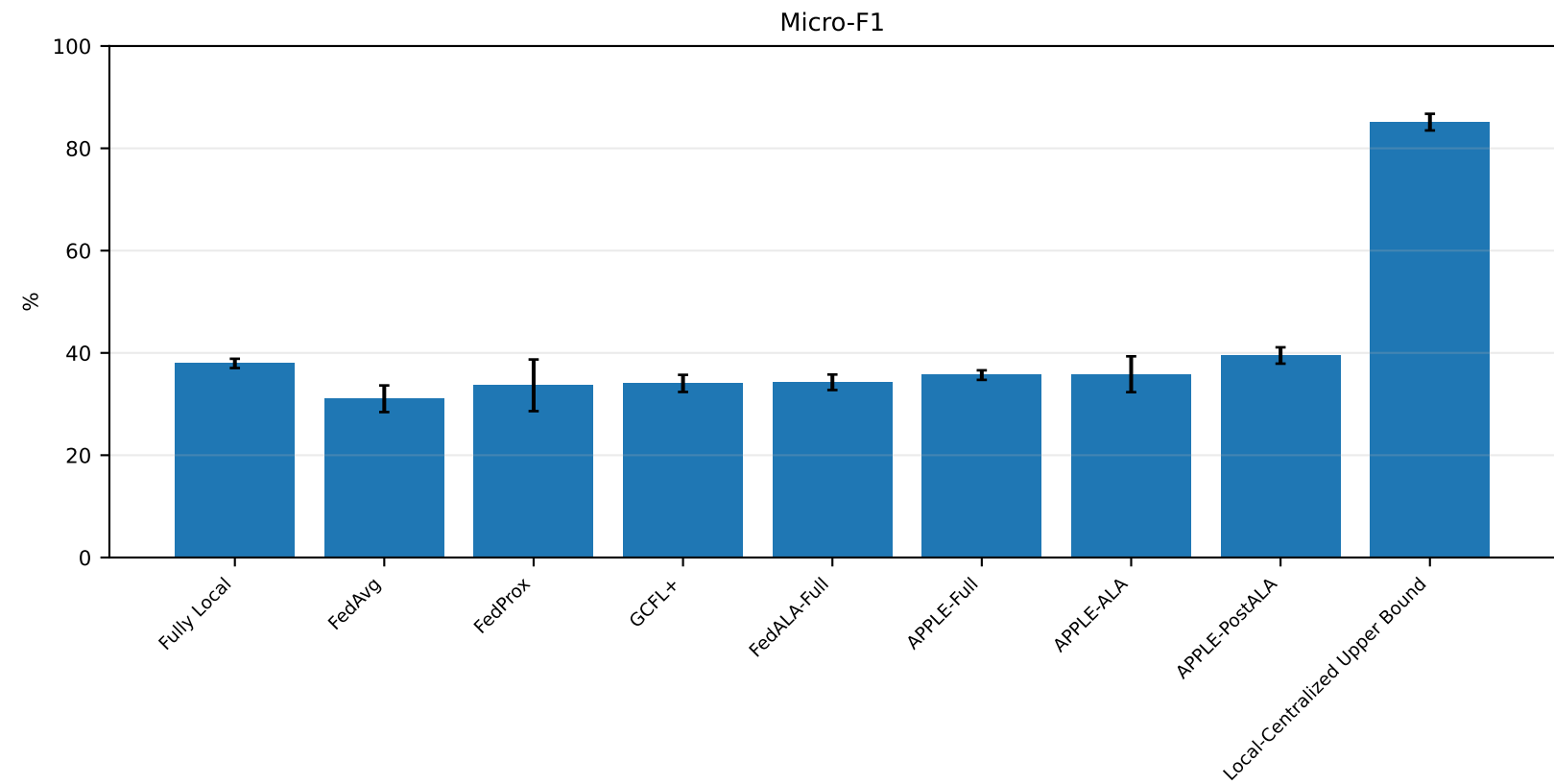
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pos_f1 | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pos_f1 | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	37.9% ± 0.9%	45.7% ± 1.2%	53.1% ± 3.6%	49.0% ± 2.2%	70.1% ± 1.2%	61.2% ± 1.5%	39.5% ± 3.4%	36.5% ± 3.6%	21.4% ± 1.1%	—	3
FedAvg	31.0% ± 2.6%	34.6% ± 2.6%	53.7% ± 0.6%	50.1% ± 1.1%	48.9% ± 7.5%	39.3% ± 5.3%	33.7% ± 2.0%	29.8% ± 5.5%	21.5% ± 2.0%	70.3 ± 12.4	3
FedProx	33.7% ± 5.1%	38.8% ± 5.2%	58.2% ± 3.3%	54.5% ± 3.6%	57.4% ± 3.0%	48.7% ± 13.2%	33.2% ± 1.5%	31.8% ± 4.6%	22.9% ± 4.7%	69.0 ± 9.5	3
GCFL+	34.0% ± 1.7%	38.8% ± 0.8%	61.4% ± 8.1%	57.7% ± 7.9%	51.4% ± 14.3%	49.6% ± 6.3%	37.6% ± 0.9%	32.6% ± 4.2%	22.5% ± 1.8%	74.3 ± 6.4	3
FedALA-Full	34.3% ± 1.5%	39.4% ± 1.6%	56.4% ± 11.9%	54.9% ± 10.6%	58.8% ± 0.9%	49.5% ± 5.0%	35.1% ± 1.8%	31.3% ± 3.6%	22.6% ± 1.1%	68.0 ± 2.6	3
APPLE-Full	35.7% ± 0.9%	41.5% ± 0.3%	49.0% ± 2.8%	45.9% ± 3.7%	64.2% ± 3.1%	53.1% ± 4.6%	31.7% ± 2.5%	34.6% ± 1.0%	24.0% ± 1.6%	73.7 ± 4.5	3
APPLE-ALA	35.8% ± 3.5%	43.7% ± 4.1%	56.1% ± 10.4%	53.8% ± 9.7%	73.6% ± 8.4%	51.4% ± 2.4%	39.5% ± 7.8%	32.9% ± 1.6%	21.1% ± 1.9%	58.0 ± 16.5	3
APPLE-PostALA	39.5% ± 1.6%	47.3% ± 3.0%	63.5% ± 6.7%	59.1% ± 7.0%	64.9% ± 5.5%	66.7% ± 7.6%	44.5% ± 6.9%	36.2% ± 3.0%	24.2% ± 2.9%	76.7 ± 3.2	3
Local-Centralized Upper Bound	85.1% ± 1.6%	87.5% ± 1.1%	95.3% ± 0.3%	93.3% ± 1.4%	96.9% ± 2.9%	98.0% ± 1.9%	89.7% ± 3.2%	83.4% ± 0.6%	69.7% ± 5.9%	—	3

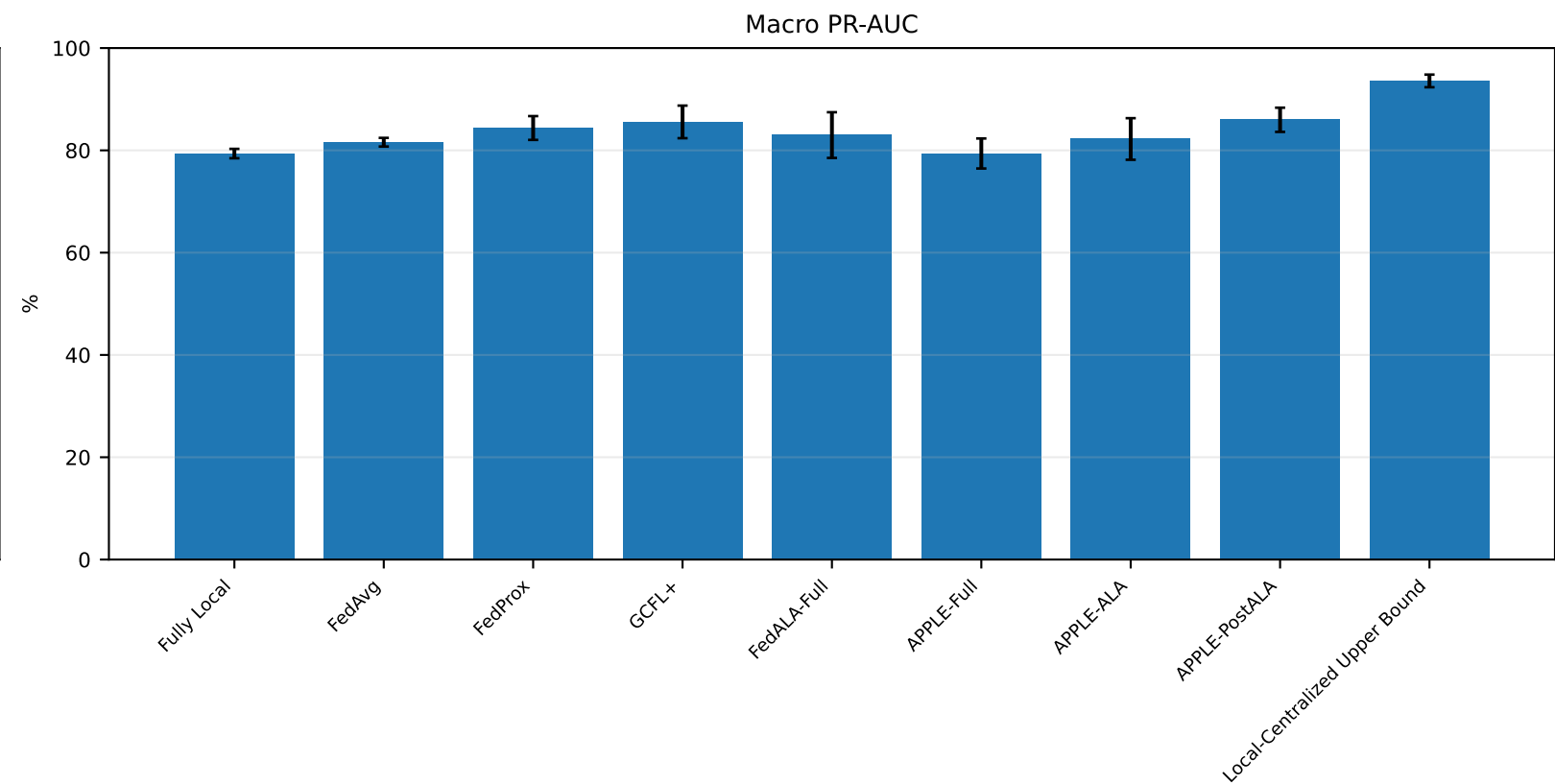
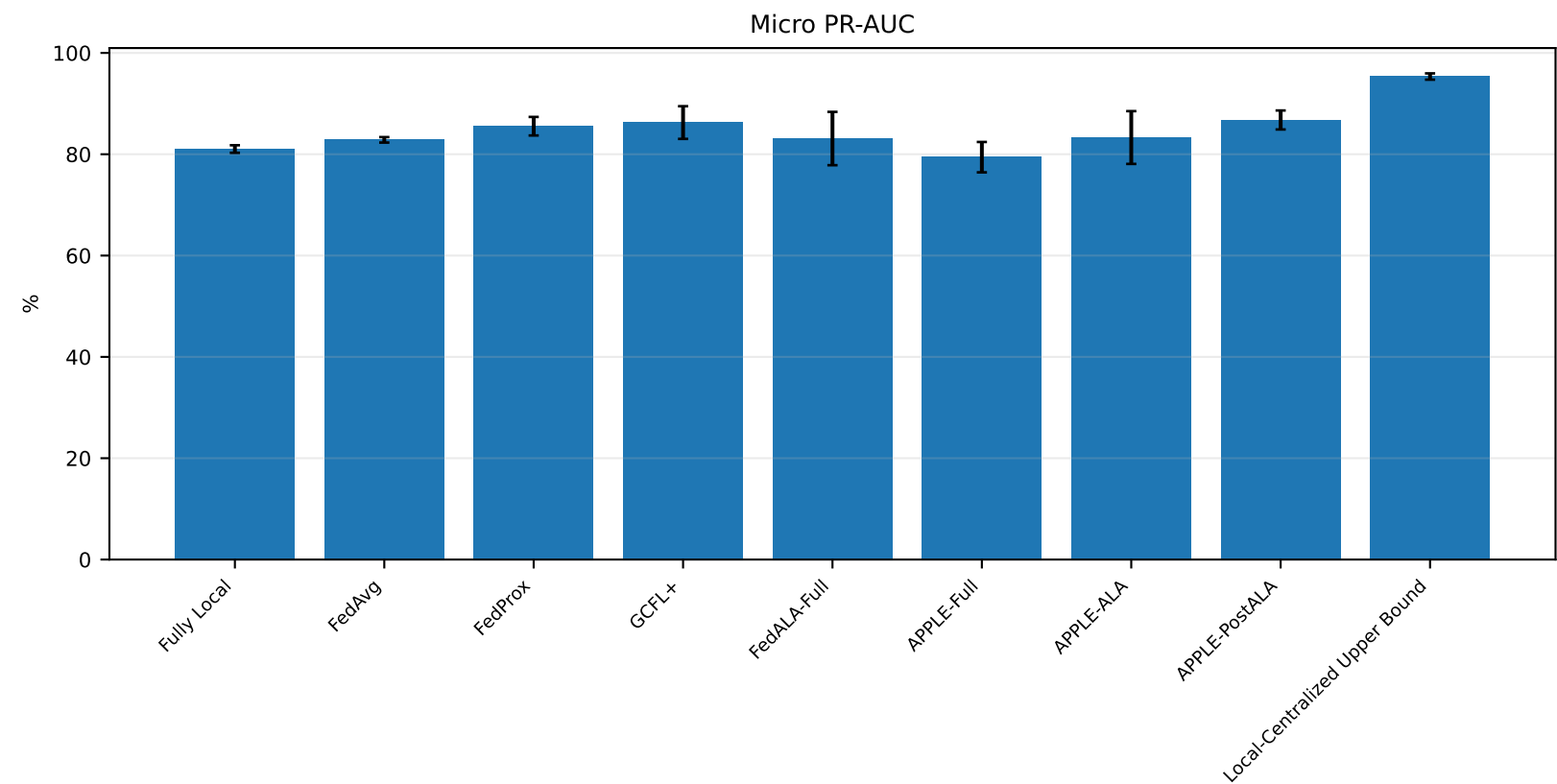
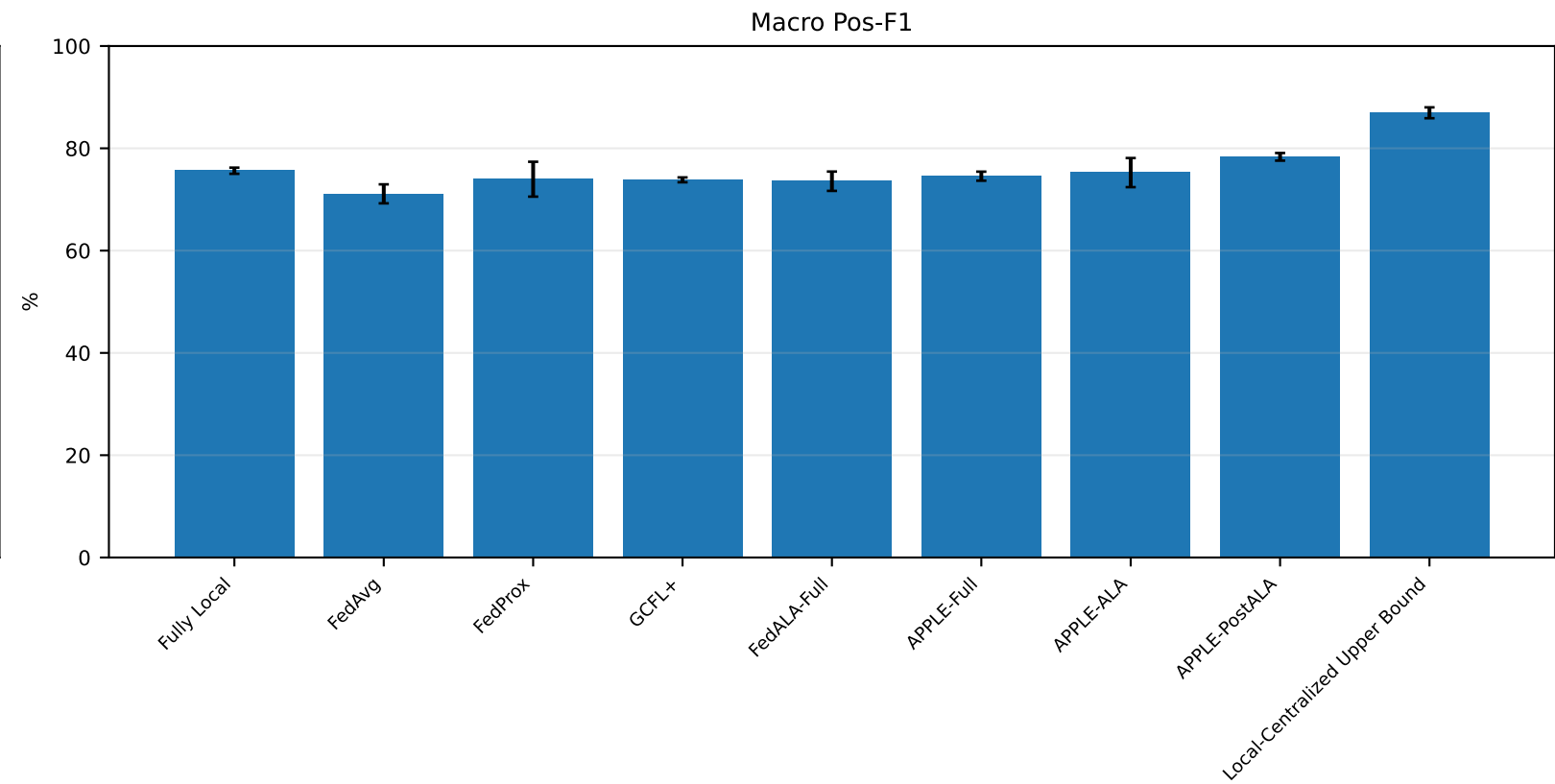
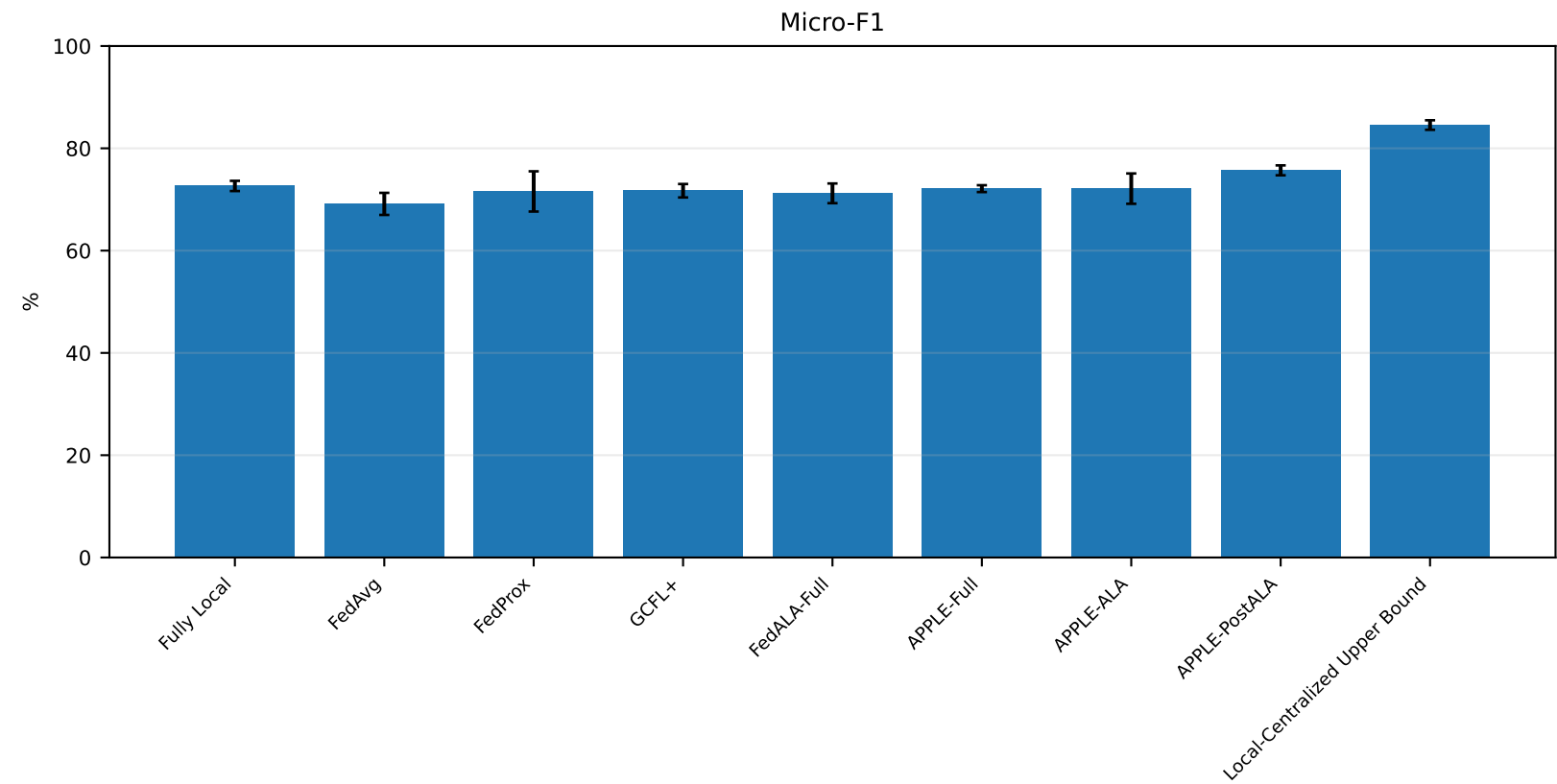
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pos_f1 | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pos_f1 | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	72.6% ± 1.0%	75.6% ± 0.6%	81.0% ± 0.7%	79.4% ± 0.9%	90.7% ± 0.3%	87.0% ± 1.7%	75.5% ± 1.1%	73.3% ± 2.9%	51.6% ± 2.3%	—	3
FedAvg	69.1% ± 2.2%	71.1% ± 1.9%	82.8% ± 0.5%	81.6% ± 0.8%	82.6% ± 4.2%	76.1% ± 3.5%	71.3% ± 1.4%	67.6% ± 5.0%	57.8% ± 2.4%	70.3 ± 12.4	3
FedProx	71.6% ± 3.9%	74.0% ± 3.4%	85.5% ± 1.8%	84.4% ± 2.3%	87.7% ± 0.8%	81.3% ± 6.3%	71.4% ± 0.9%	69.9% ± 3.7%	59.5% ± 5.9%	69.0 ± 9.5	3
GCFL+	71.7% ± 1.3%	73.8% ± 0.5%	86.3% ± 3.2%	85.6% ± 3.2%	82.8% ± 7.0%	82.7% ± 3.0%	74.4% ± 0.3%	70.4% ± 4.0%	58.9% ± 2.3%	74.3 ± 6.4	3
FedALA-Full	71.2% ± 1.9%	73.6% ± 1.9%	83.1% ± 5.3%	83.0% ± 4.5%	86.9% ± 2.2%	81.4% ± 3.5%	72.1% ± 2.2%	69.0% ± 3.3%	58.4% ± 1.7%	68.0 ± 2.6	3
APPLE-Full	72.1% ± 0.7%	74.6% ± 0.9%	79.4% ± 3.0%	79.4% ± 2.9%	88.3% ± 2.2%	84.1% ± 2.7%	68.9% ± 2.3%	71.9% ± 0.6%	59.5% ± 1.3%	73.7 ± 4.5	3
APPLE-ALA	72.1% ± 3.0%	75.3% ± 2.8%	83.3% ± 5.2%	82.2% ± 4.1%	92.1% ± 3.0%	83.2% ± 2.6%	74.9% ± 5.9%	69.9% ± 1.4%	56.2% ± 2.5%	58.0 ± 16.5	3
APPLE-PostALA	75.7% ± 1.0%	78.3% ± 0.7%	86.8% ± 1.9%	86.0% ± 2.4%	89.2% ± 1.8%	90.8% ± 2.6%	77.4% ± 1.7%	73.4% ± 2.1%	60.9% ± 4.0%	76.7 ± 3.2	3
Local-Centralized Upper Bound	84.6% ± 0.9%	86.9% ± 1.1%	95.3% ± 0.6%	93.6% ± 1.2%	96.1% ± 3.8%	96.9% ± 2.2%	89.3% ± 3.2%	83.5% ± 2.2%	69.0% ± 5.1%	—	3

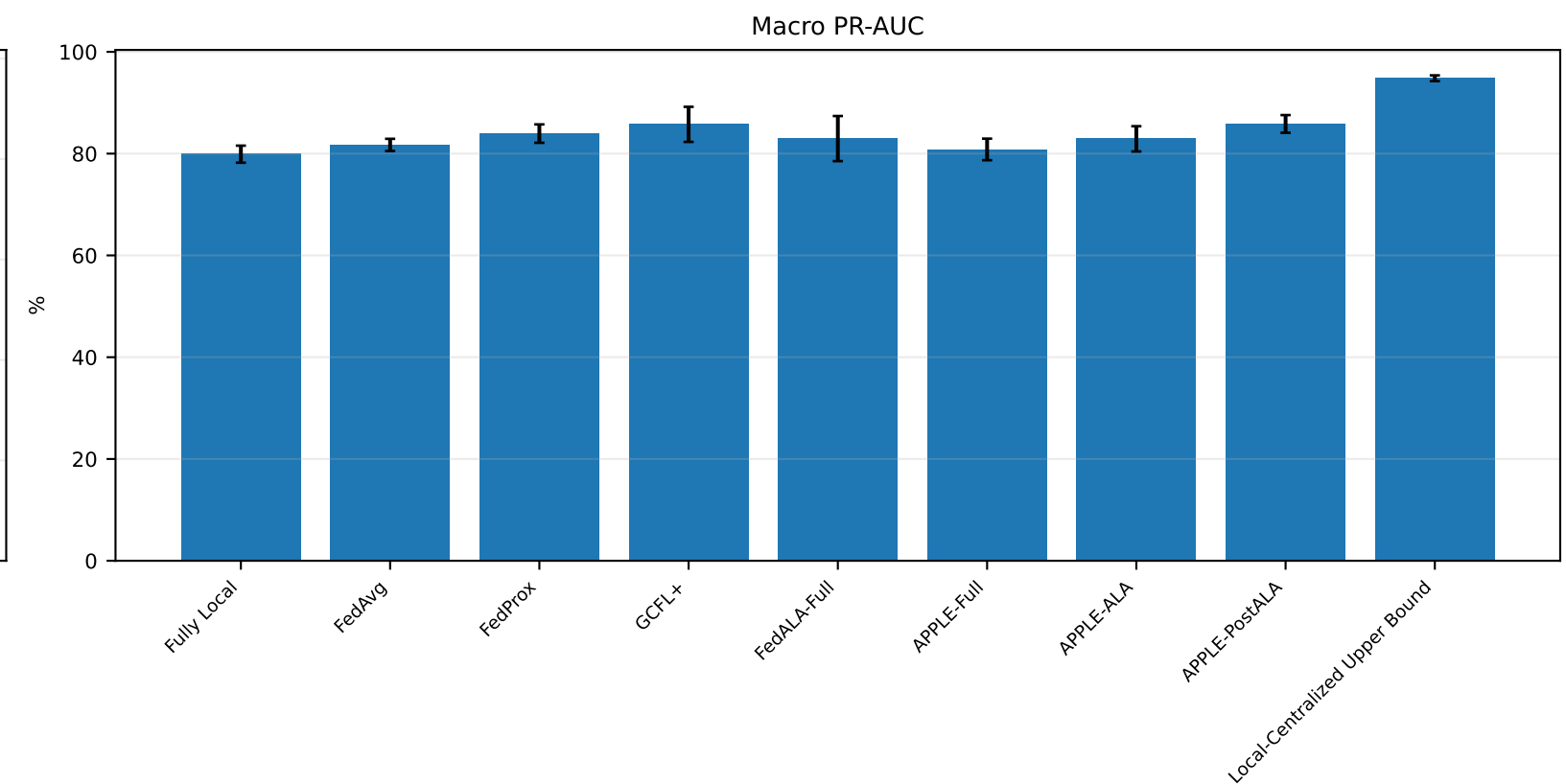
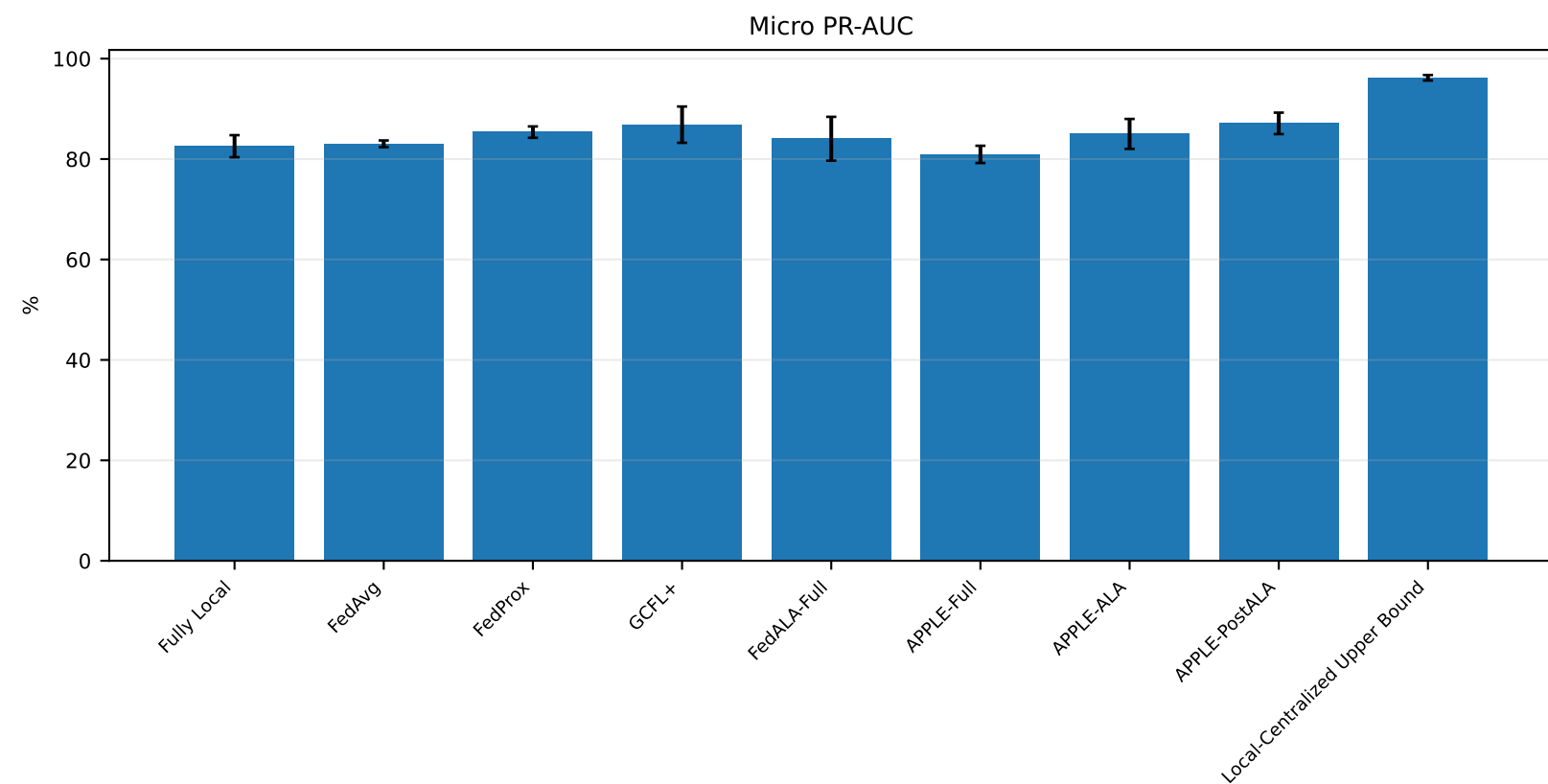
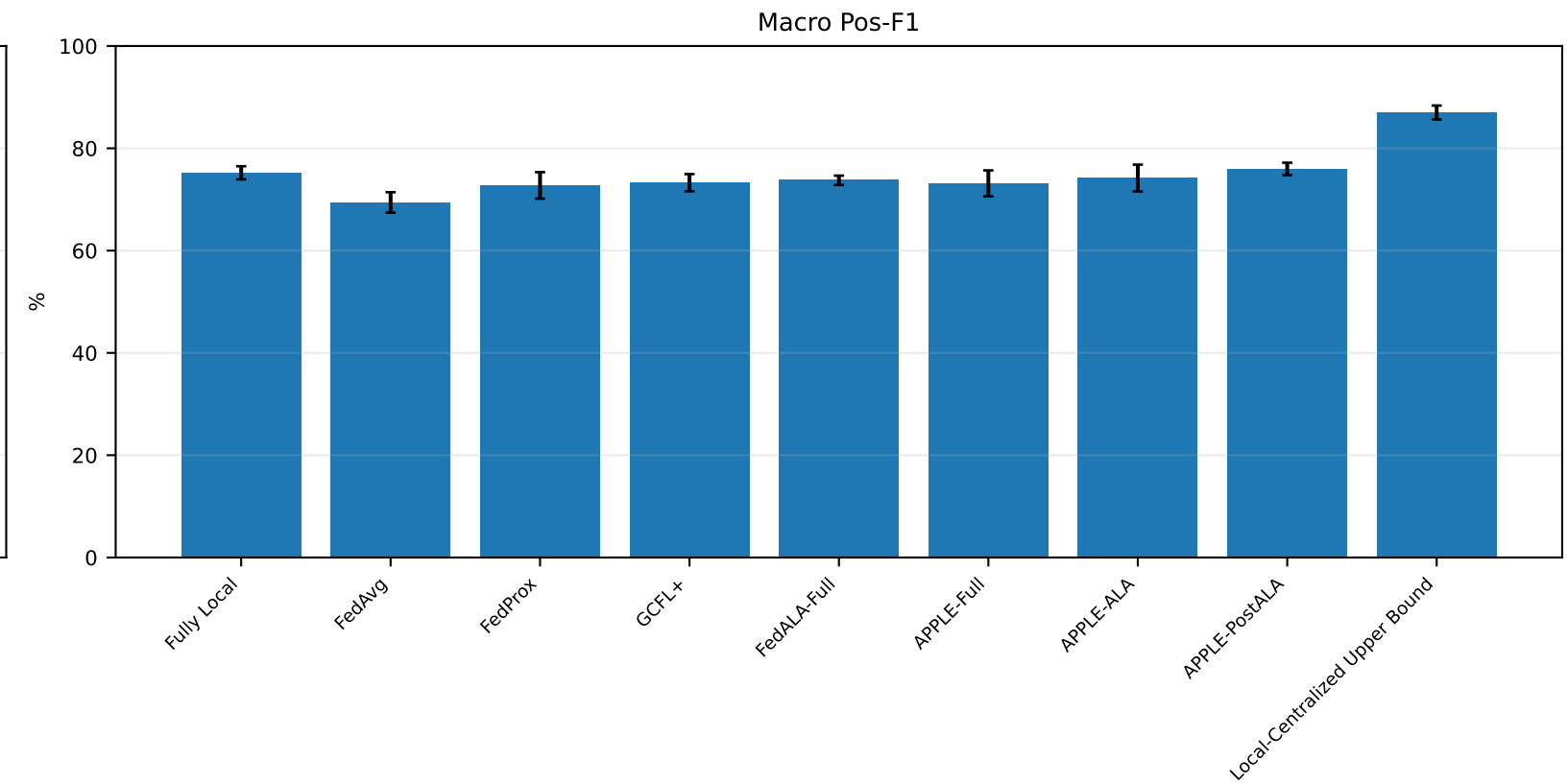
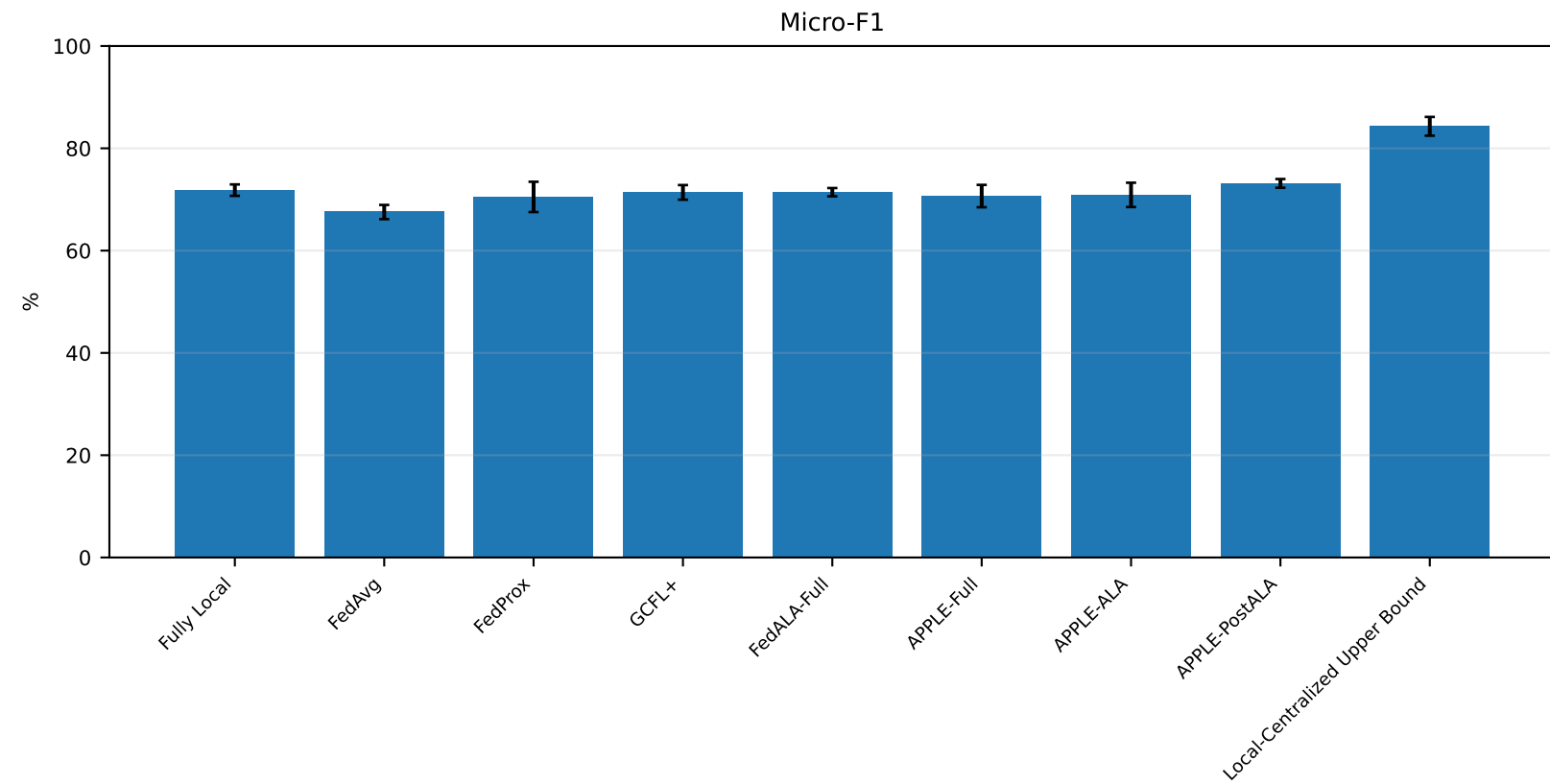
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pos_f1 | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	71.8% ± 1.1%	75.2% ± 1.3%	82.6% ± 2.2%	79.9% ± 1.7%	91.0% ± 1.7%	87.8% ± 1.4%	73.4% ± 1.8%	72.0% ± 2.4%	51.8% ± 2.0%	—	3
FedAvg	67.5% ± 1.4%	69.4% ± 2.0%	83.0% ± 0.7%	81.7% ± 1.2%	79.2% ± 5.7%	77.2% ± 0.9%	68.2% ± 4.0%	65.7% ± 2.2%	56.8% ± 1.4%	69.3 ± 11.7	3
FedProx	70.5% ± 3.0%	72.7% ± 2.6%	85.4% ± 1.1%	83.9% ± 1.8%	84.5% ± 3.1%	81.5% ± 4.3%	69.8% ± 0.9%	69.3% ± 3.9%	58.6% ± 4.5%	79.3 ± 1.2	3
GCFL+	71.4% ± 1.4%	73.3% ± 1.7%	86.8% ± 3.6%	85.8% ± 3.4%	81.0% ± 5.8%	83.5% ± 0.8%	72.6% ± 2.5%	69.6% ± 3.3%	59.6% ± 0.7%	67.3 ± 9.9	3
FedALA-Full	71.4% ± 0.8%	73.7% ± 0.9%	84.0% ± 4.4%	83.0% ± 4.4%	88.0% ± 0.4%	80.5% ± 4.6%	73.2% ± 0.8%	68.4% ± 1.2%	58.6% ± 1.5%	66.0 ± 9.6	3
APPLE-Full	70.7% ± 2.2%	73.2% ± 2.5%	80.9% ± 1.7%	80.8% ± 2.1%	85.8% ± 6.7%	82.9% ± 1.6%	67.2% ± 3.9%	71.8% ± 0.8%	58.1% ± 1.8%	69.3 ± 11.0	3
APPLE-ALA	70.9% ± 2.4%	74.2% ± 2.6%	85.0% ± 3.0%	82.9% ± 2.5%	91.5% ± 4.0%	82.5% ± 4.1%	74.6% ± 3.0%	67.2% ± 0.5%	55.1% ± 2.5%	74.3 ± 5.7	3
APPLE-PostALA	73.1% ± 0.9%	76.0% ± 1.2%	87.1% ± 2.1%	85.8% ± 1.7%	88.6% ± 2.7%	89.0% ± 2.9%	75.2% ± 3.1%	69.2% ± 1.2%	57.9% ± 2.1%	74.7 ± 2.9	3
Local-Centralized Upper Bound	84.3% ± 1.8%	87.0% ± 1.4%	96.2% ± 0.5%	94.8% ± 0.6%	98.7% ± 0.6%	97.3% ± 0.9%	88.4% ± 1.2%	82.3% ± 4.0%	68.3% ± 3.3%	—	3

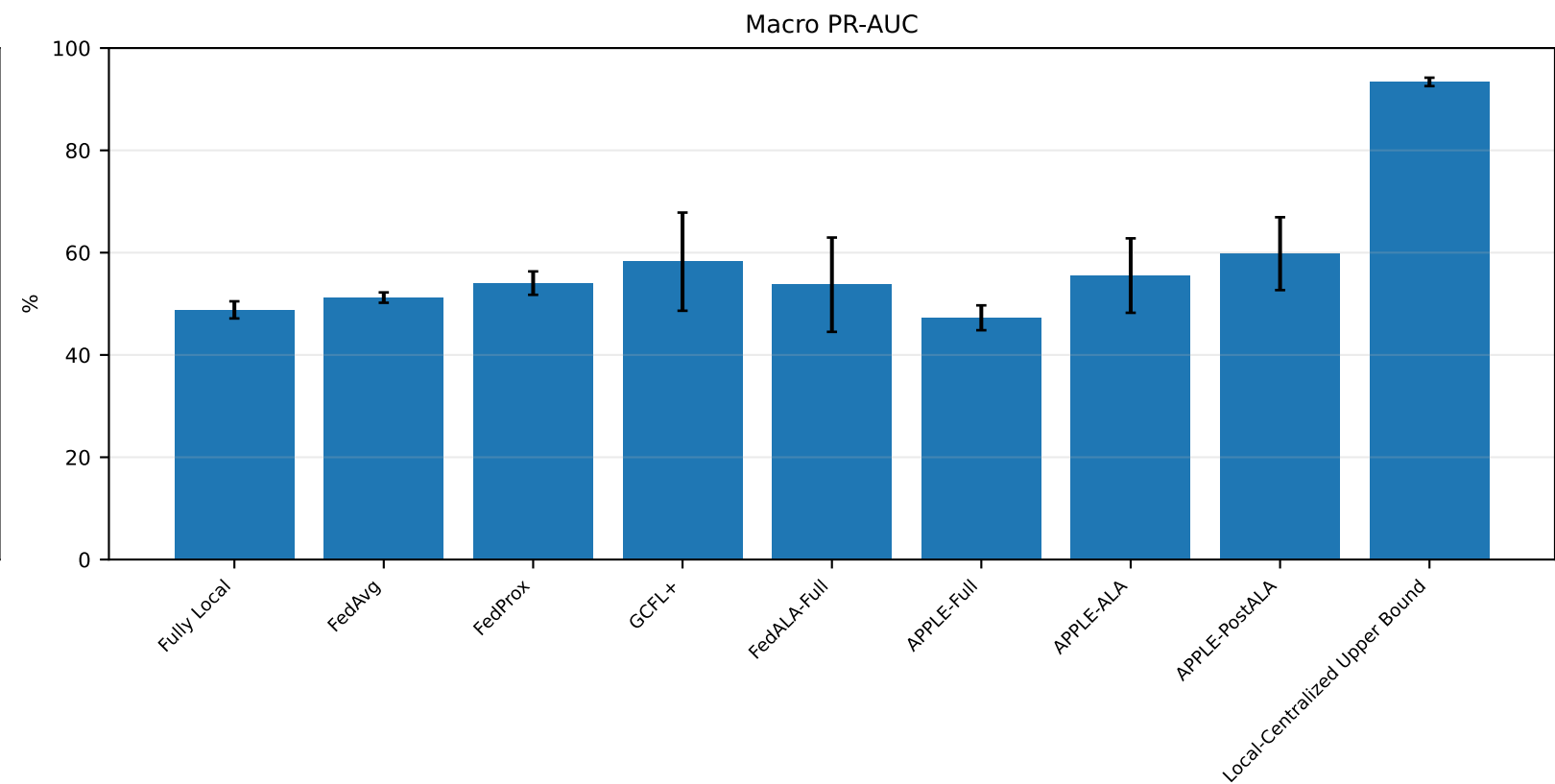
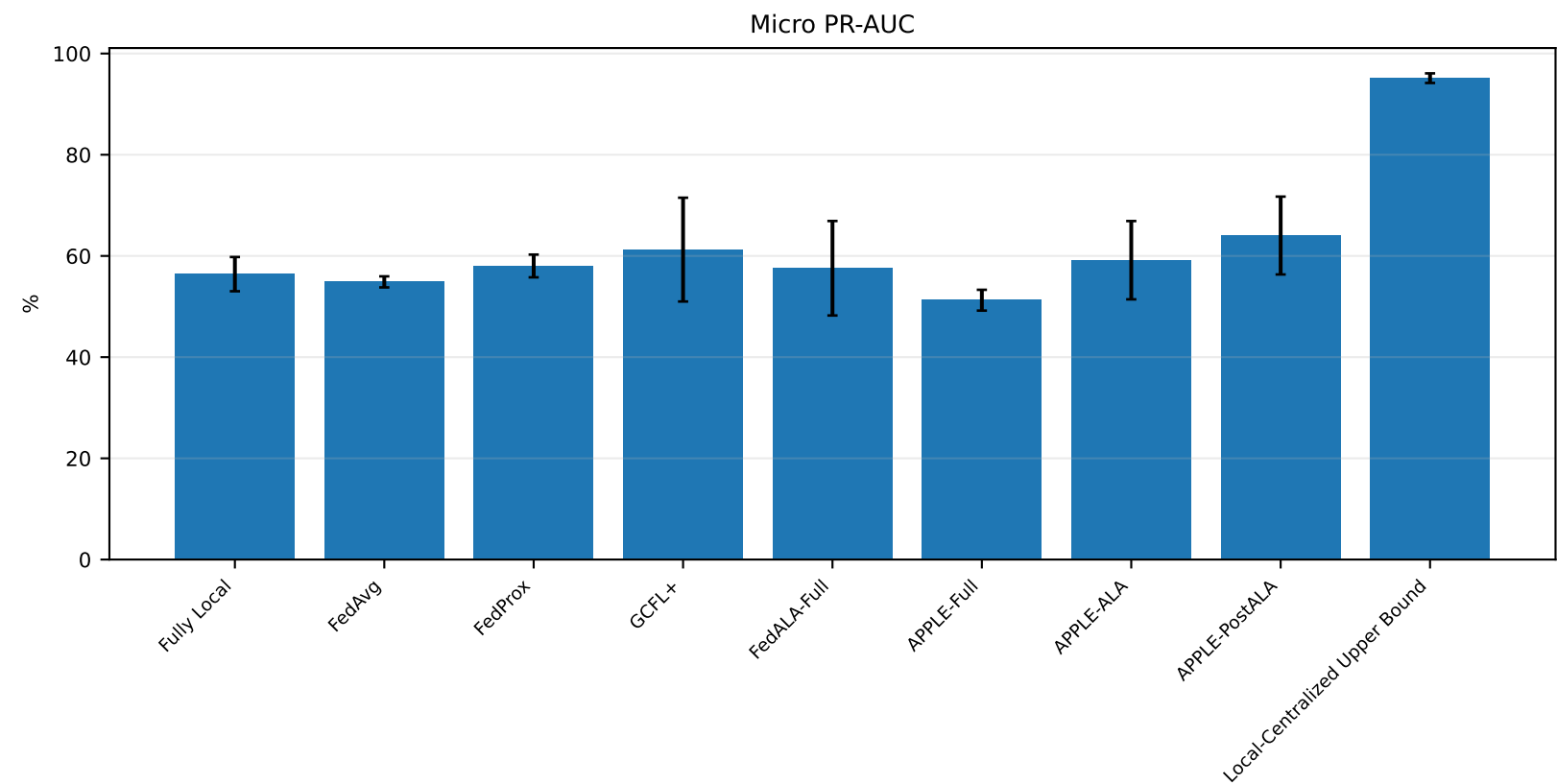
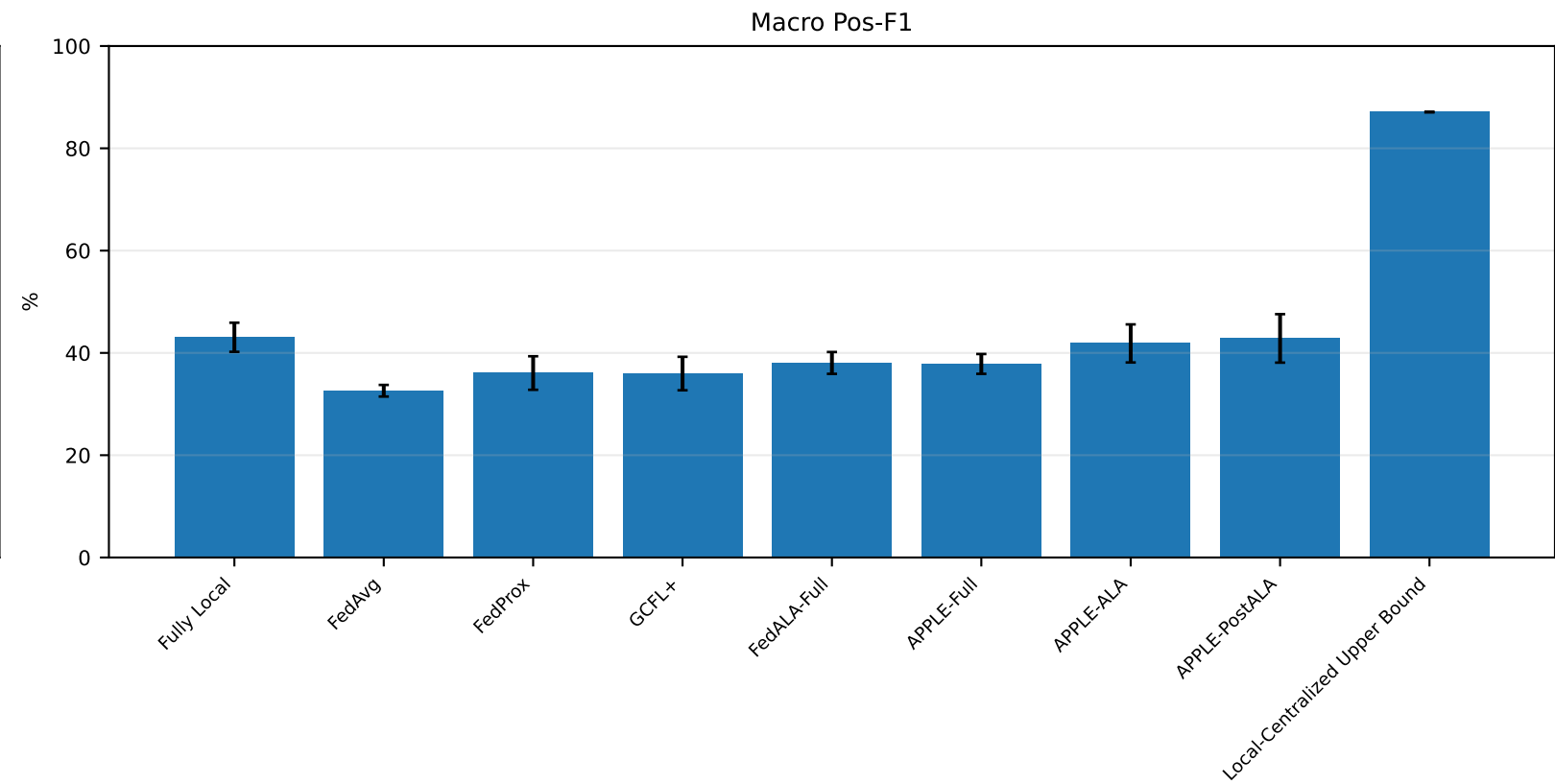
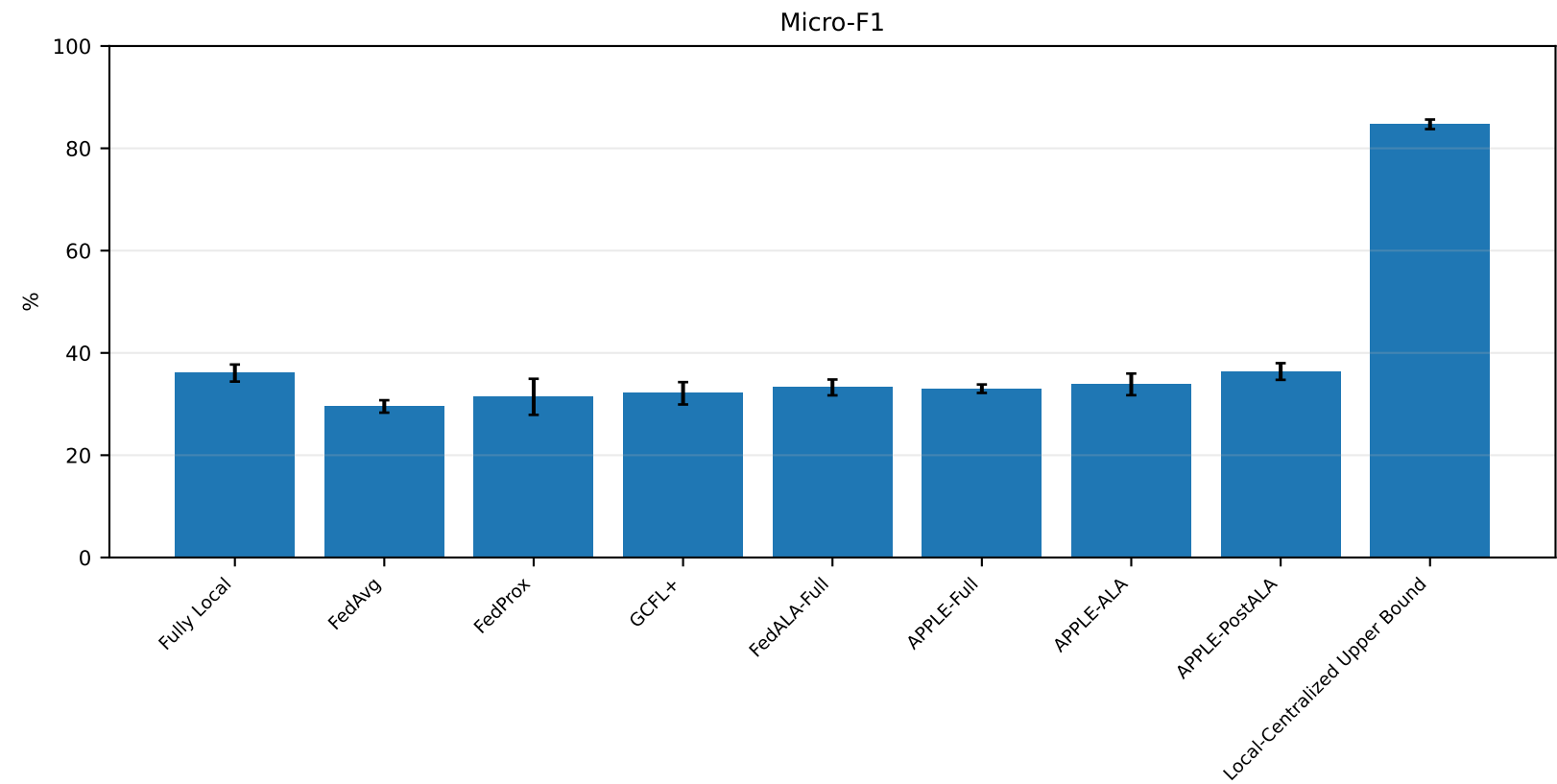
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	36.1% ± 1.7%	43.1% ± 2.8%	56.4% ± 3.4%	48.8% ± 1.7%	64.7% ± 3.9%	57.0% ± 6.1%	38.1% ± 4.1%	34.4% ± 2.8%	21.1% ± 1.2%	—	3
FedAvg	29.5% ± 1.2%	32.6% ± 1.1%	54.9% ± 1.1%	51.2% ± 1.0%	44.6% ± 8.0%	37.0% ± 3.5%	30.4% ± 1.8%	30.1% ± 4.4%	20.9% ± 1.9%	71.3 ± 9.6	3
FedProx	31.4% ± 3.5%	36.1% ± 3.3%	58.0% ± 2.2%	54.0% ± 2.3%	52.1% ± 3.0%	46.2% ± 8.4%	30.6% ± 2.5%	30.1% ± 4.7%	21.3% ± 2.9%	74.3 ± 6.0	3
GCFL+	32.1% ± 2.2%	36.0% ± 3.3%	61.2% ± 10.3%	58.2% ± 9.6%	45.6% ± 11.0%	48.1% ± 4.5%	33.4% ± 3.9%	30.7% ± 2.8%	22.1% ± 1.5%	74.3 ± 5.5	3
FedALA-Full	33.3% ± 1.5%	38.0% ± 2.1%	57.6% ± 9.3%	53.7% ± 9.2%	57.8% ± 1.2%	46.3% ± 7.0%	32.7% ± 3.8%	31.2% ± 4.0%	22.3% ± 1.3%	75.7 ± 4.9	3
APPLE-Full	33.0% ± 0.8%	37.8% ± 1.9%	51.3% ± 2.1%	47.3% ± 2.4%	55.5% ± 7.2%	49.0% ± 4.6%	29.0% ± 2.1%	33.1% ± 1.2%	22.6% ± 1.0%	67.3 ± 14.8	3
APPLE-ALA	33.9% ± 2.1%	41.9% ± 3.7%	59.2% ± 7.7%	55.5% ± 7.3%	72.1% ± 7.9%	48.4% ± 7.5%	38.2% ± 4.4%	30.2% ± 1.4%	20.5% ± 1.6%	71.3 ± 10.8	3
APPLE-PostALA	36.4% ± 1.6%	42.8% ± 4.7%	64.0% ± 7.7%	59.8% ± 7.1%	60.9% ± 9.3%	61.0% ± 11.6%	38.0% ± 5.8%	31.7% ± 2.0%	22.6% ± 2.7%	75.7 ± 2.5	3
Local-Centralized Upper Bound	84.7% ± 0.9%	87.1% ± 0.1%	95.1% ± 0.9%	93.4% ± 0.8%	97.2% ± 2.5%	96.1% ± 1.2%	90.1% ± 2.5%	82.9% ± 2.6%	69.2% ± 5.6%	—	3

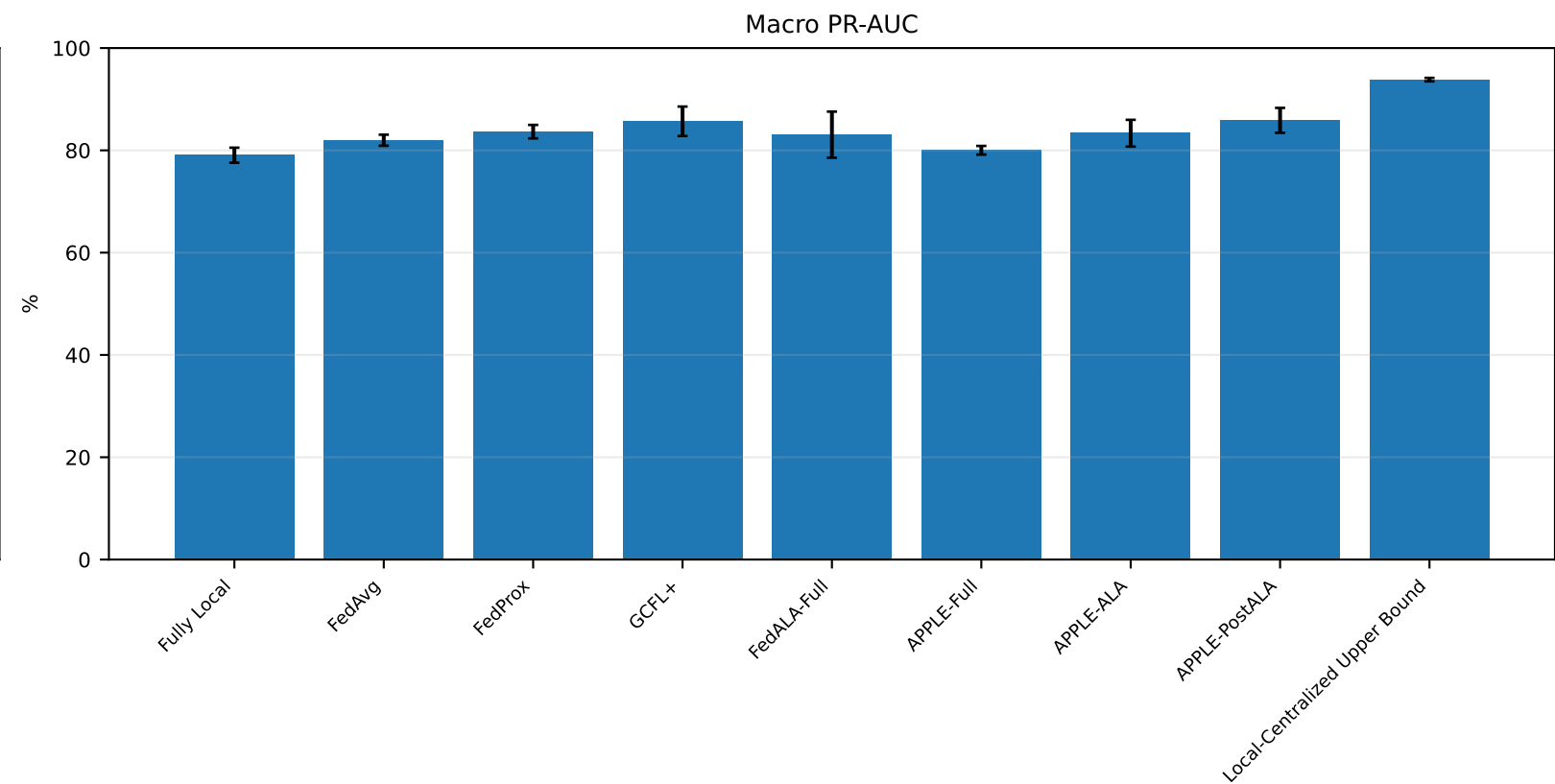
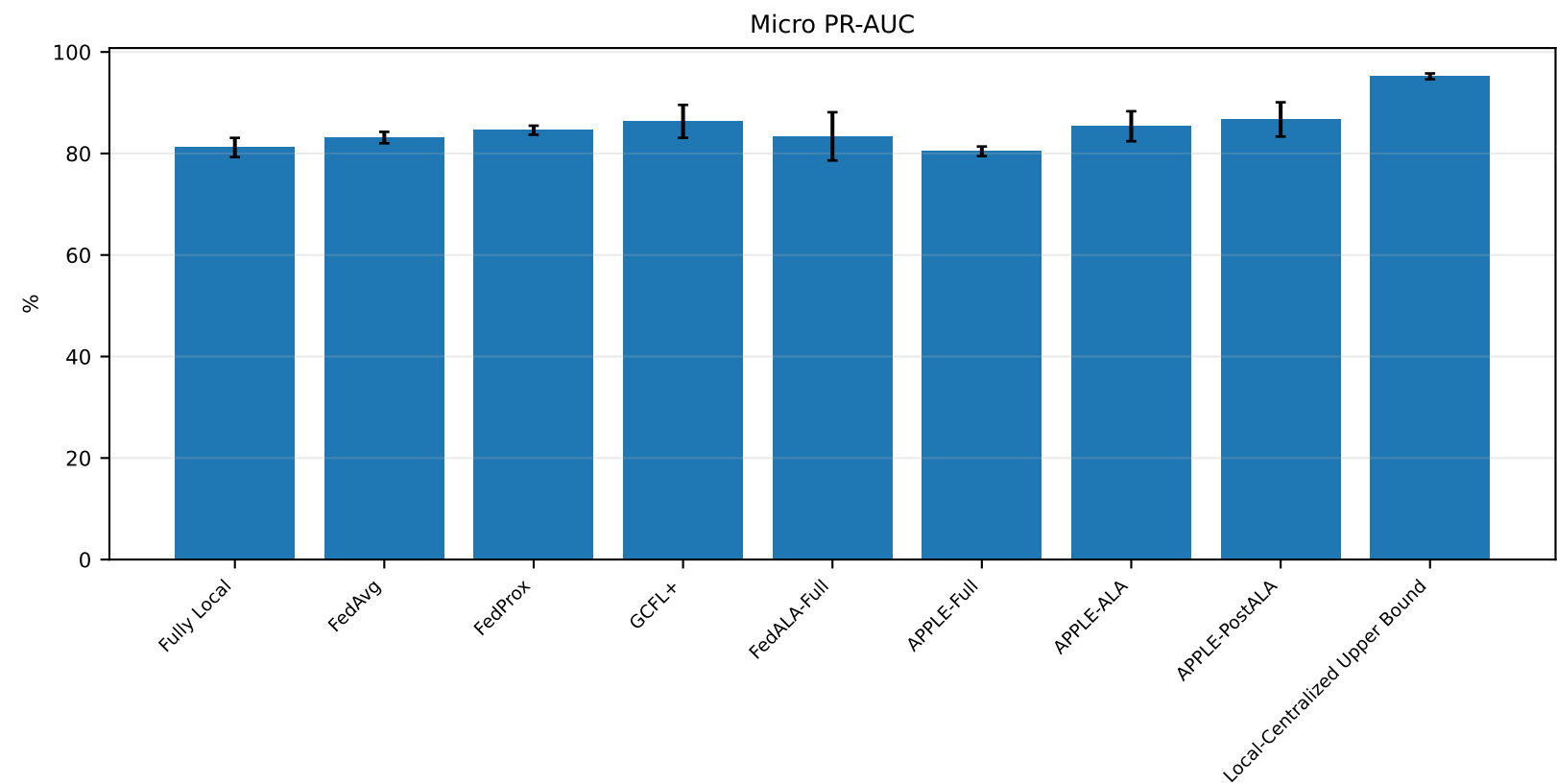
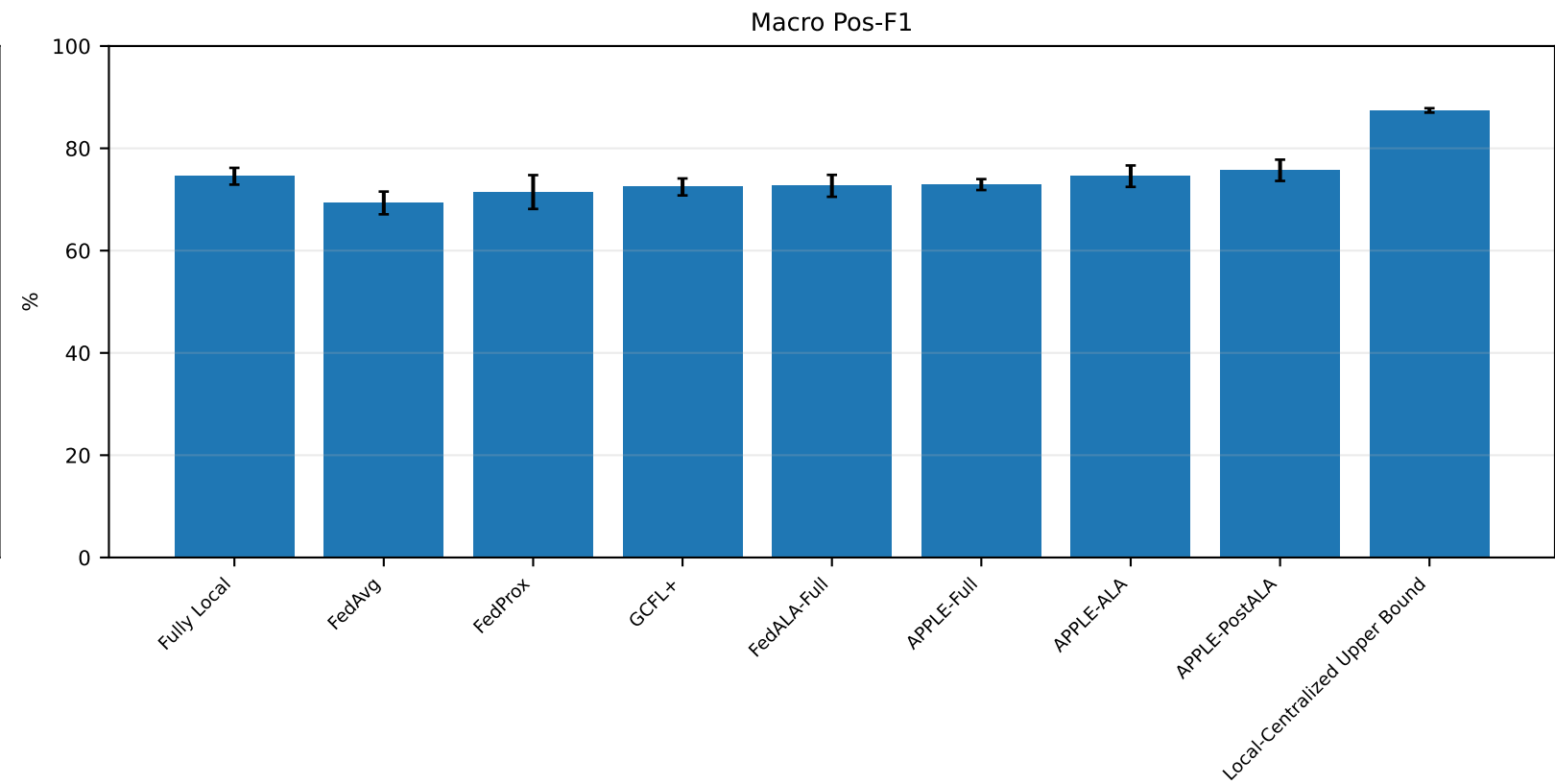
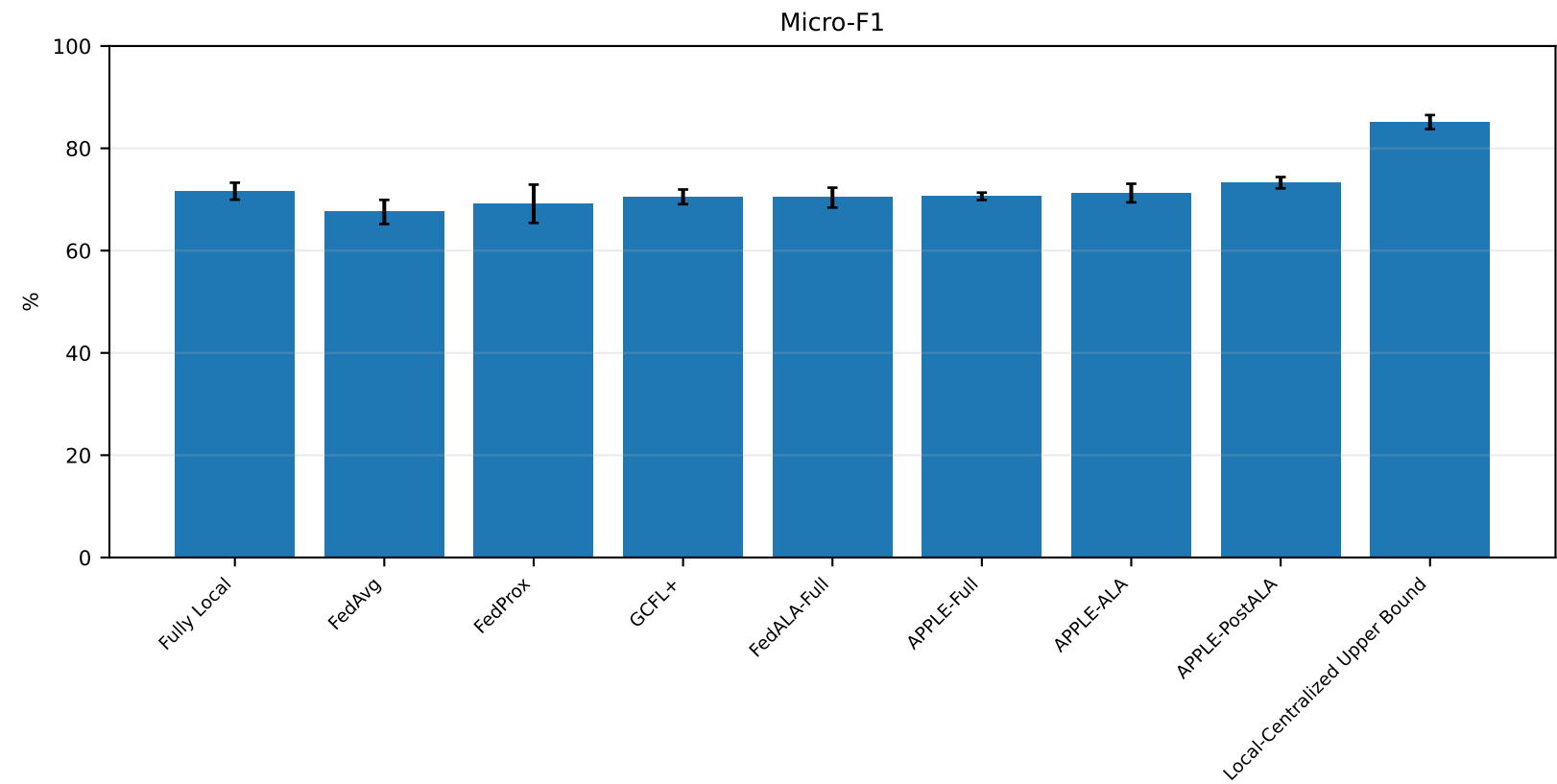
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=micro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	71.6% ± 1.7%	74.5% ± 1.6%	81.2% ± 1.9%	79.1% ± 1.5%	88.3% ± 2.0%	87.1% ± 2.3%	73.6% ± 2.4%	71.9% ± 3.3%	51.9% ± 2.2%	—	3
FedAvg	67.6% ± 2.4%	69.3% ± 2.2%	83.1% ± 1.1%	82.0% ± 1.1%	79.1% ± 5.4%	74.2% ± 1.8%	68.5% ± 2.6%	67.8% ± 5.3%	56.9% ± 3.6%	71.3 ± 9.6	3
FedProx	69.2% ± 3.7%	71.5% ± 3.3%	84.6% ± 0.9%	83.7% ± 1.3%	83.9% ± 2.9%	80.0% ± 4.8%	68.3% ± 2.0%	67.6% ± 4.9%	57.5% ± 4.8%	74.3 ± 6.0	3
GCFL+	70.5% ± 1.4%	72.4% ± 1.7%	86.3% ± 3.2%	85.7% ± 2.9%	80.1% ± 6.1%	82.6% ± 2.8%	71.5% ± 3.2%	69.4% ± 2.2%	58.6% ± 1.4%	74.3 ± 5.5	3
FedALA-Full	70.4% ± 1.9%	72.7% ± 2.1%	83.4% ± 4.8%	83.1% ± 4.5%	86.8% ± 1.5%	79.5% ± 5.0%	70.3% ± 4.8%	68.7% ± 3.2%	58.0% ± 1.8%	75.7 ± 4.9	3
APPLE-Full	70.6% ± 0.7%	72.9% ± 1.1%	80.4% ± 0.9%	80.0% ± 0.8%	85.7% ± 3.8%	82.5% ± 2.3%	66.6% ± 1.6%	71.4% ± 0.4%	58.3% ± 1.3%	67.3 ± 14.8	3
APPLE-ALA	71.3% ± 1.8%	74.6% ± 2.1%	85.4% ± 3.0%	83.4% ± 2.6%	92.4% ± 2.6%	82.3% ± 4.3%	74.5% ± 3.1%	68.0% ± 1.2%	55.6% ± 1.9%	71.3 ± 10.8	3
APPLE-PostALA	73.3% ± 1.1%	75.7% ± 2.1%	86.7% ± 3.4%	85.9% ± 2.4%	87.5% ± 4.6%	87.5% ± 4.4%	74.8% ± 4.7%	69.6% ± 1.2%	59.0% ± 3.6%	75.7 ± 2.5	3
Local-Centralized Upper Bound	85.1% ± 1.4%	87.4% ± 0.4%	95.2% ± 0.6%	93.8% ± 0.3%	96.2% ± 2.3%	97.1% ± 1.3%	90.8% ± 1.9%	82.8% ± 2.9%	70.2% ± 5.7%	—	3

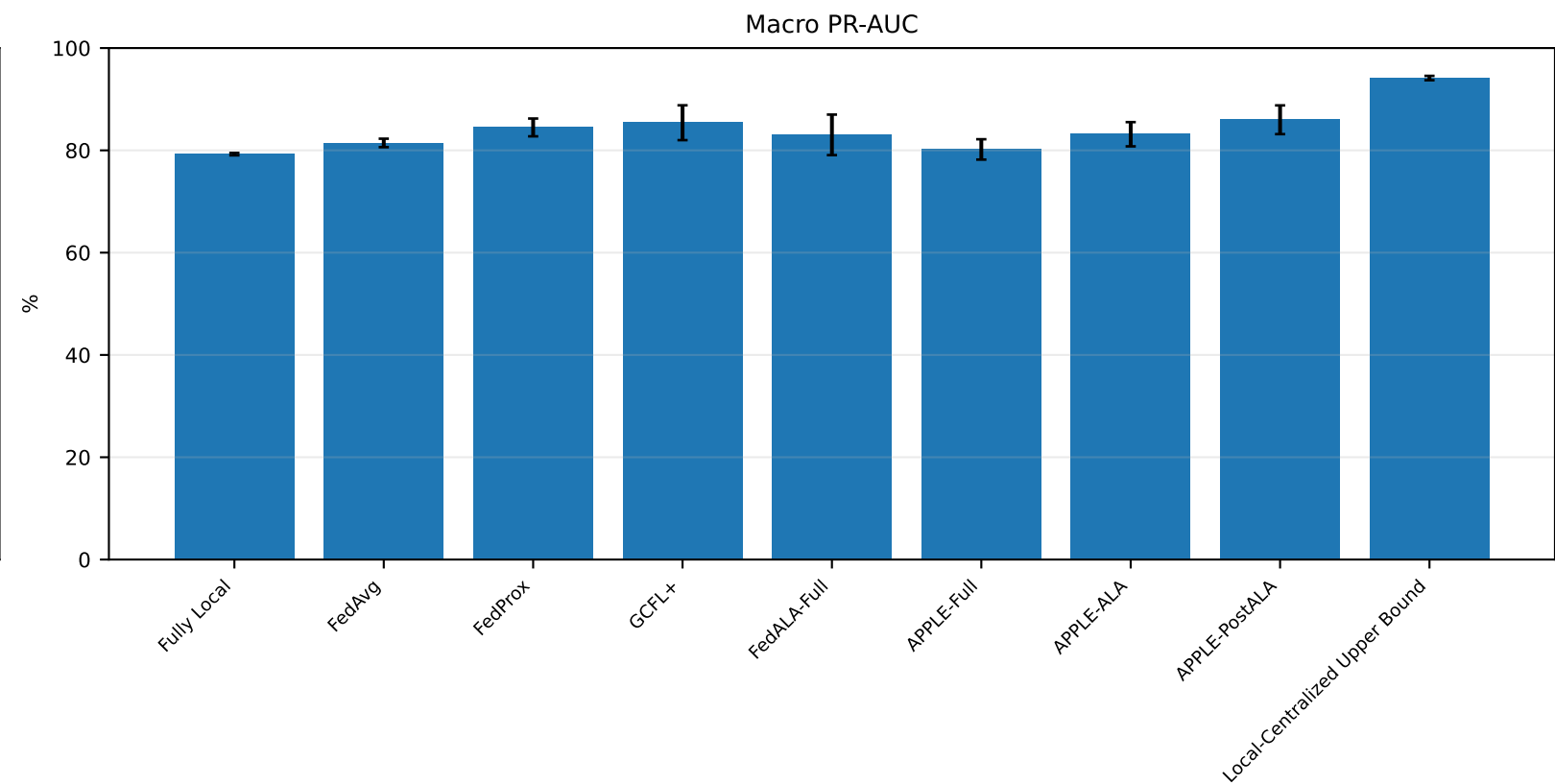
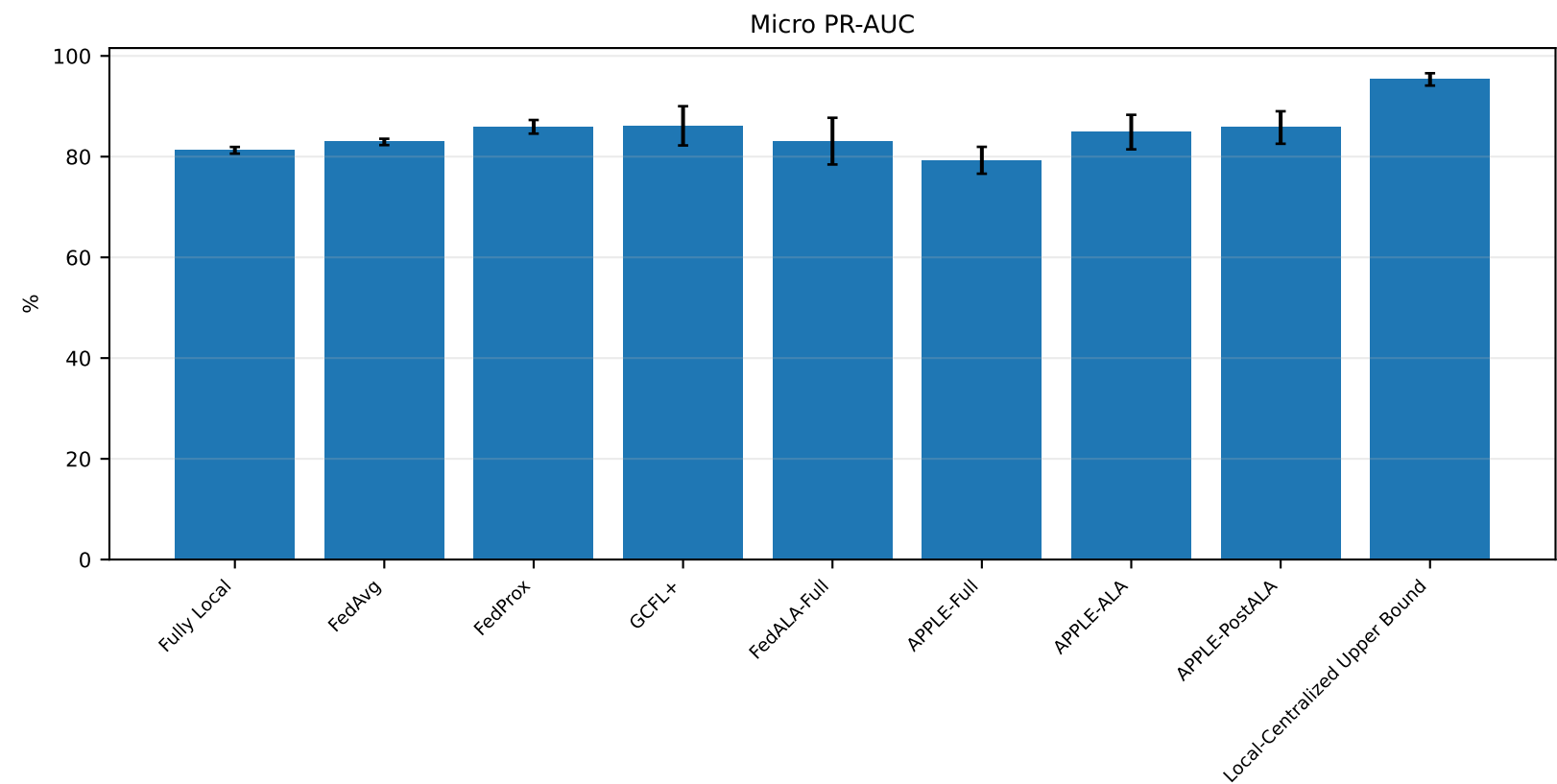
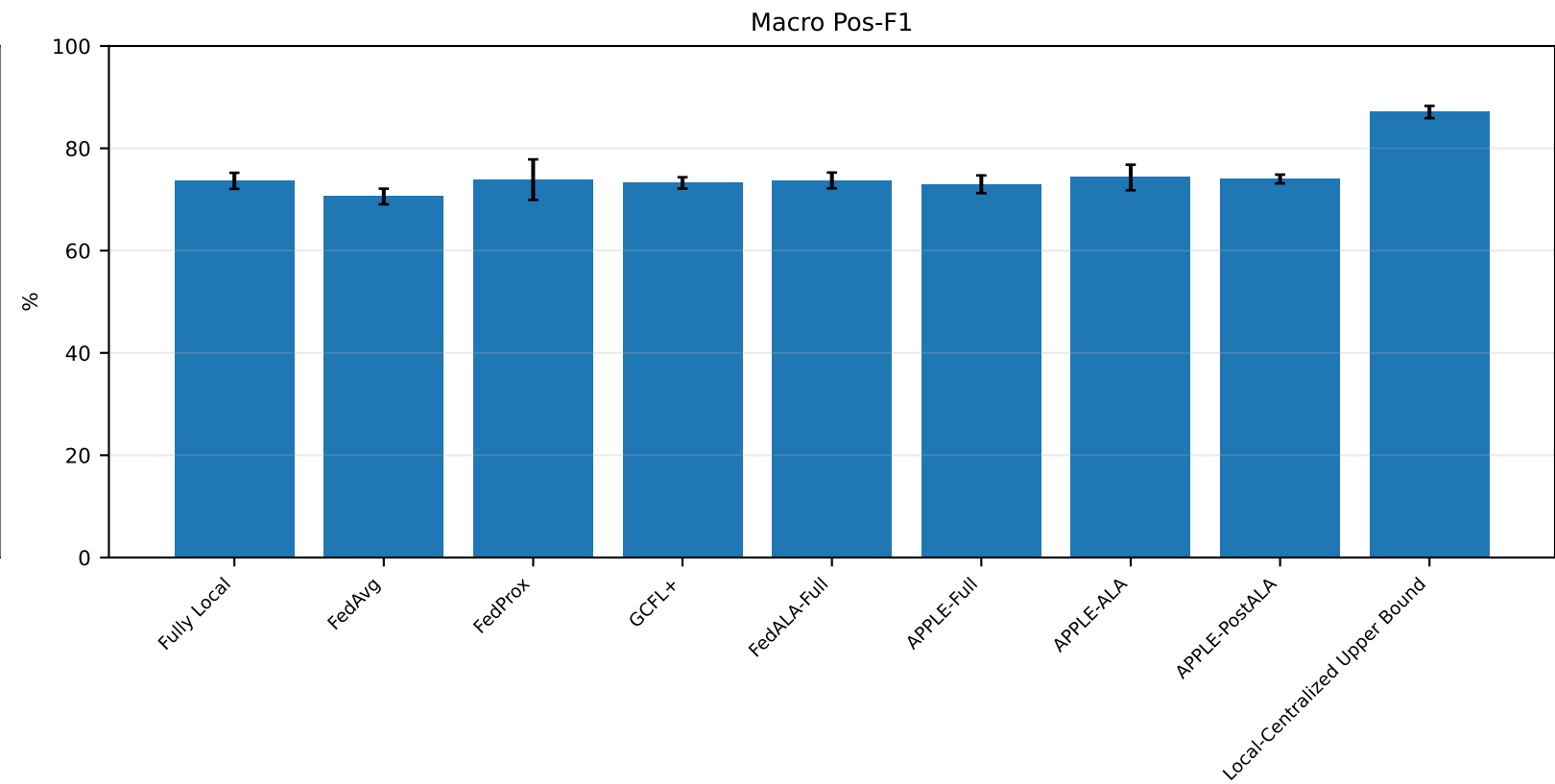
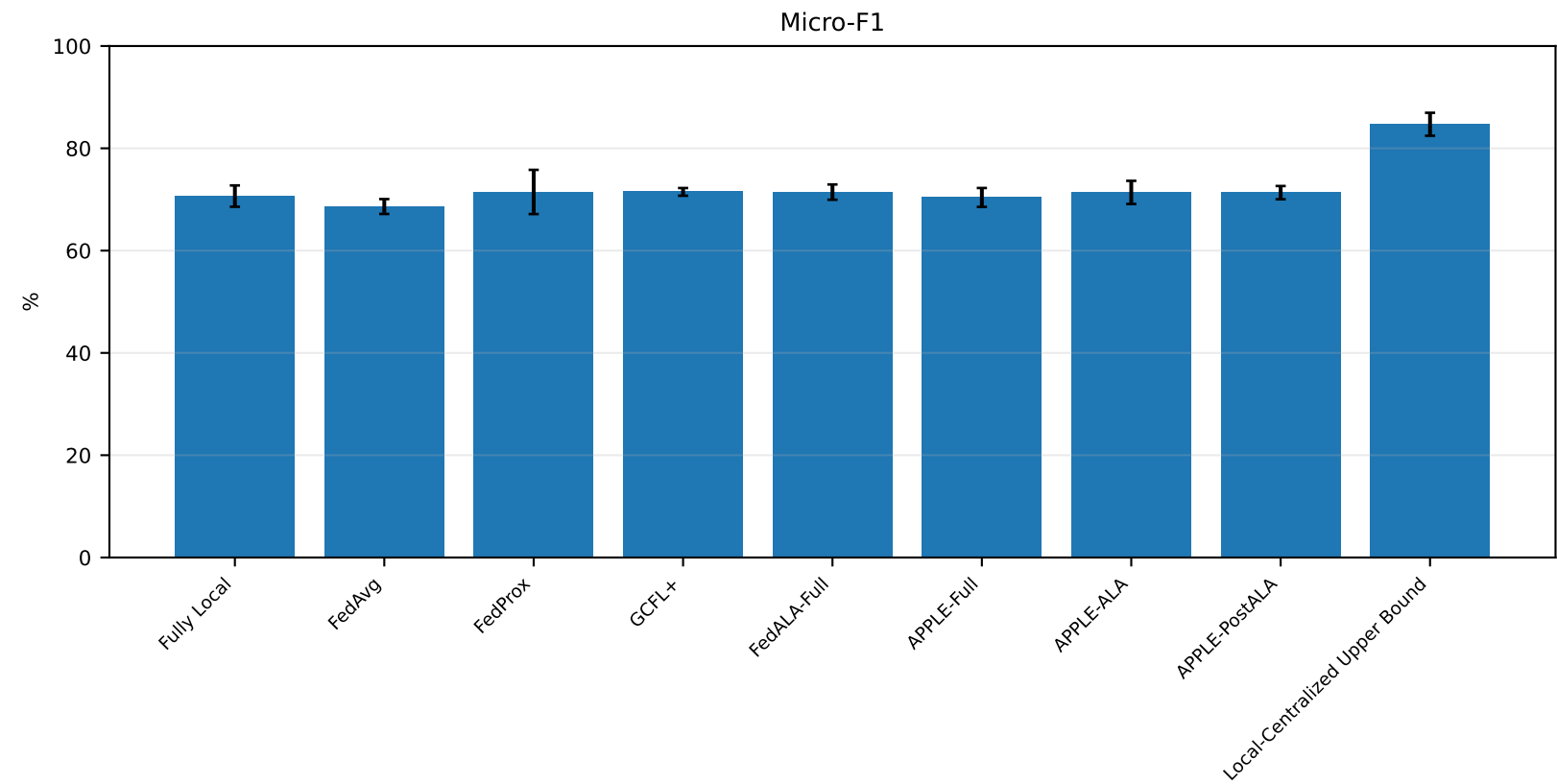
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=micro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pr_auc | protocol=OO
O+O: oracle/full validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	70.7% ± 2.1%	73.6% ± 1.6%	81.3% ± 0.7%	79.3% ± 0.2%	87.5% ± 1.8%	86.4% ± 0.5%	71.7% ± 2.2%	70.3% ± 2.1%	52.3% ± 3.3%	—	3
FedAvg	68.6% ± 1.5%	70.6% ± 1.5%	82.9% ± 0.6%	81.4% ± 0.8%	81.9% ± 6.6%	75.9% ± 3.2%	70.3% ± 2.2%	67.1% ± 4.5%	57.7% ± 2.5%	70.0 ± 12.1	3
FedProx	71.5% ± 4.3%	73.9% ± 4.0%	85.9% ± 1.4%	84.5% ± 1.7%	87.0% ± 3.6%	82.8% ± 5.6%	71.0% ± 1.8%	69.3% ± 4.1%	59.3% ± 5.9%	78.7 ± 1.2	3
GCFL+	71.5% ± 0.8%	73.2% ± 1.1%	86.1% ± 3.9%	85.4% ± 3.4%	80.2% ± 5.6%	82.6% ± 2.6%	72.3% ± 3.6%	71.3% ± 2.5%	59.9% ± 0.1%	67.0 ± 12.1	3
FedALA-Full	71.4% ± 1.5%	73.7% ± 1.5%	83.1% ± 4.6%	83.0% ± 4.0%	86.9% ± 1.7%	80.5% ± 5.1%	73.5% ± 1.1%	68.7% ± 1.7%	59.0% ± 1.4%	63.3 ± 7.6	3
APPLE-Full	70.4% ± 1.8%	73.0% ± 1.7%	79.3% ± 2.7%	80.2% ± 2.0%	86.8% ± 3.5%	83.7% ± 1.6%	67.3% ± 3.9%	68.9% ± 1.6%	58.2% ± 1.8%	66.0 ± 14.4	3
APPLE-ALA	71.4% ± 2.3%	74.3% ± 2.5%	84.9% ± 3.4%	83.2% ± 2.4%	90.2% ± 4.2%	84.6% ± 3.2%	73.4% ± 2.6%	67.3% ± 0.5%	56.0% ± 2.1%	74.3 ± 6.7	3
APPLE-PostALA	71.3% ± 1.3%	74.0% ± 0.9%	85.8% ± 3.2%	86.0% ± 2.8%	84.5% ± 0.8%	86.1% ± 1.4%	72.9% ± 1.3%	69.4% ± 2.2%	57.1% ± 3.5%	76.0 ± 5.3	3
Local-Centralized Upper Bound	84.7% ± 2.2%	87.1% ± 1.2%	95.3% ± 1.2%	94.1% ± 0.4%	97.4% ± 1.1%	95.3% ± 3.0%	90.7% ± 1.3%	82.2% ± 3.4%	69.8% ± 6.7%	—	3

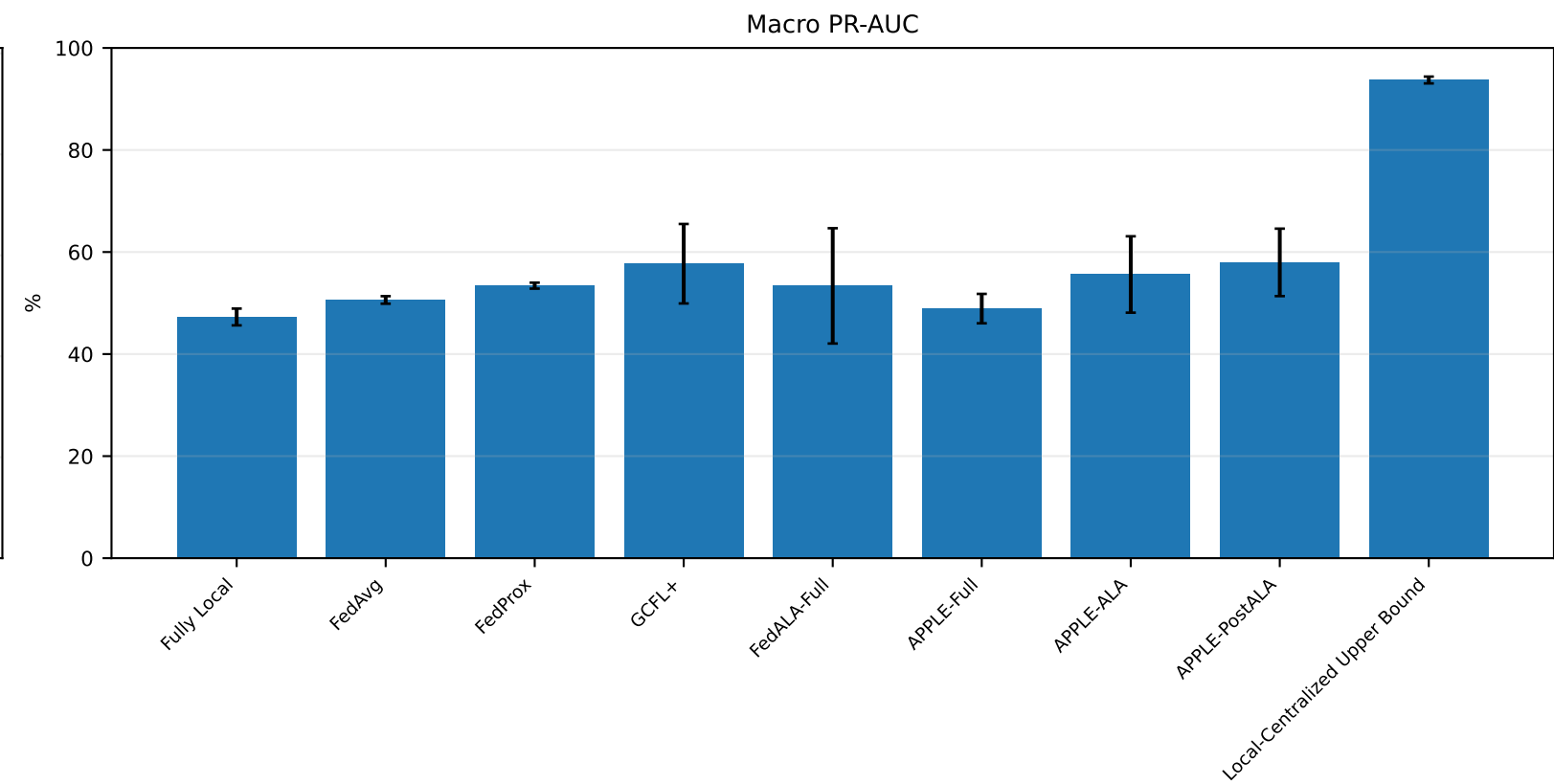
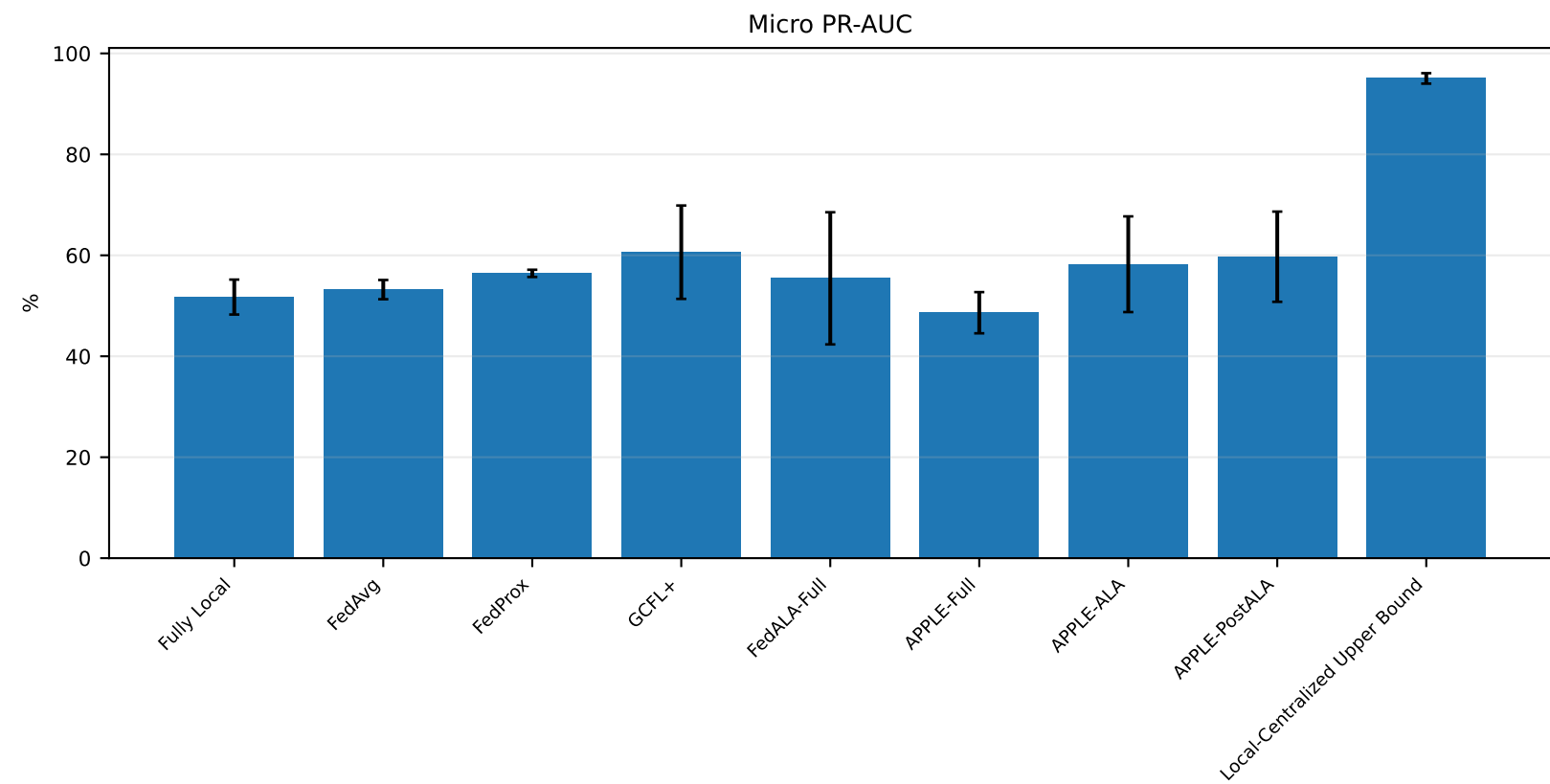
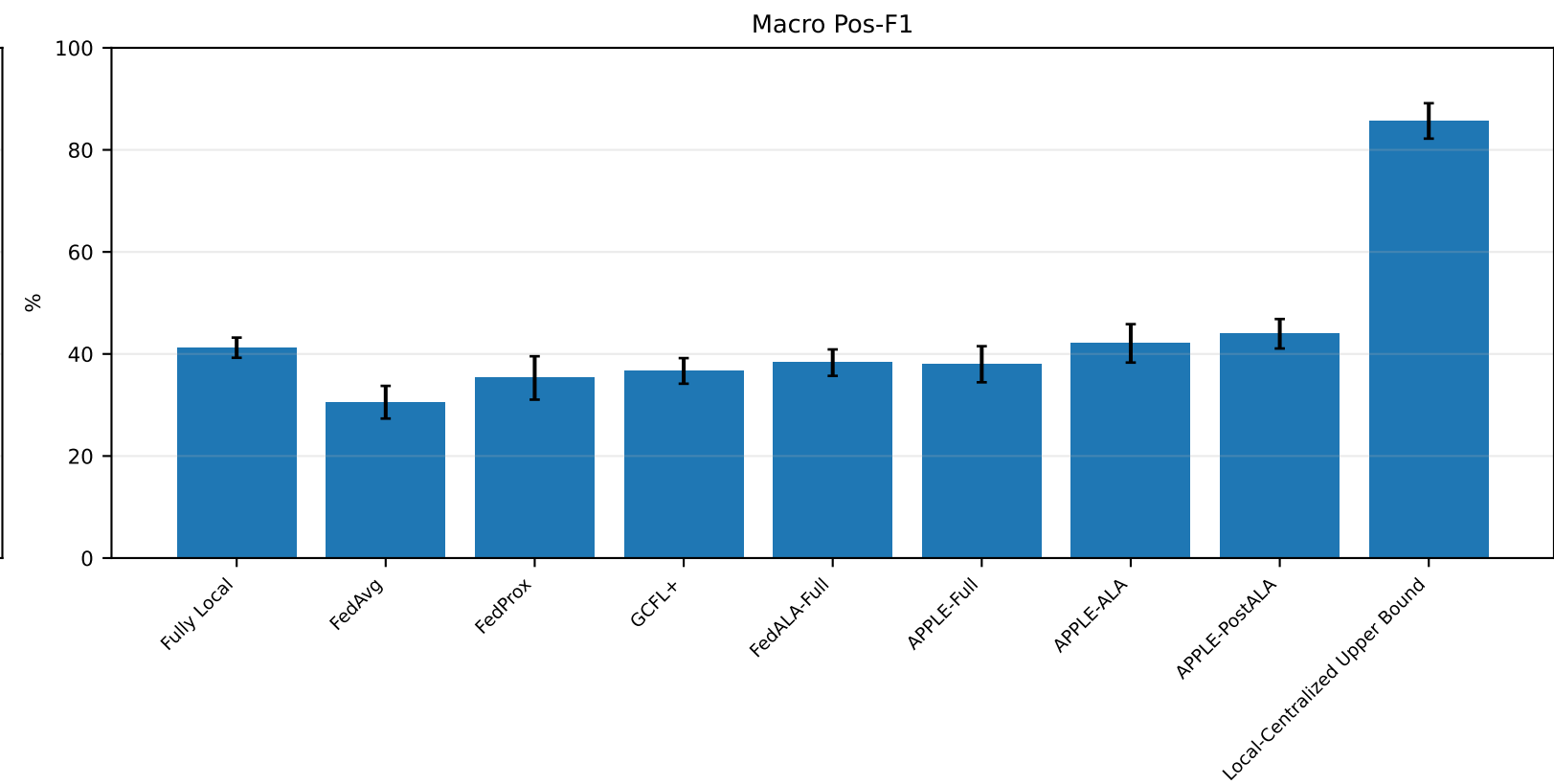
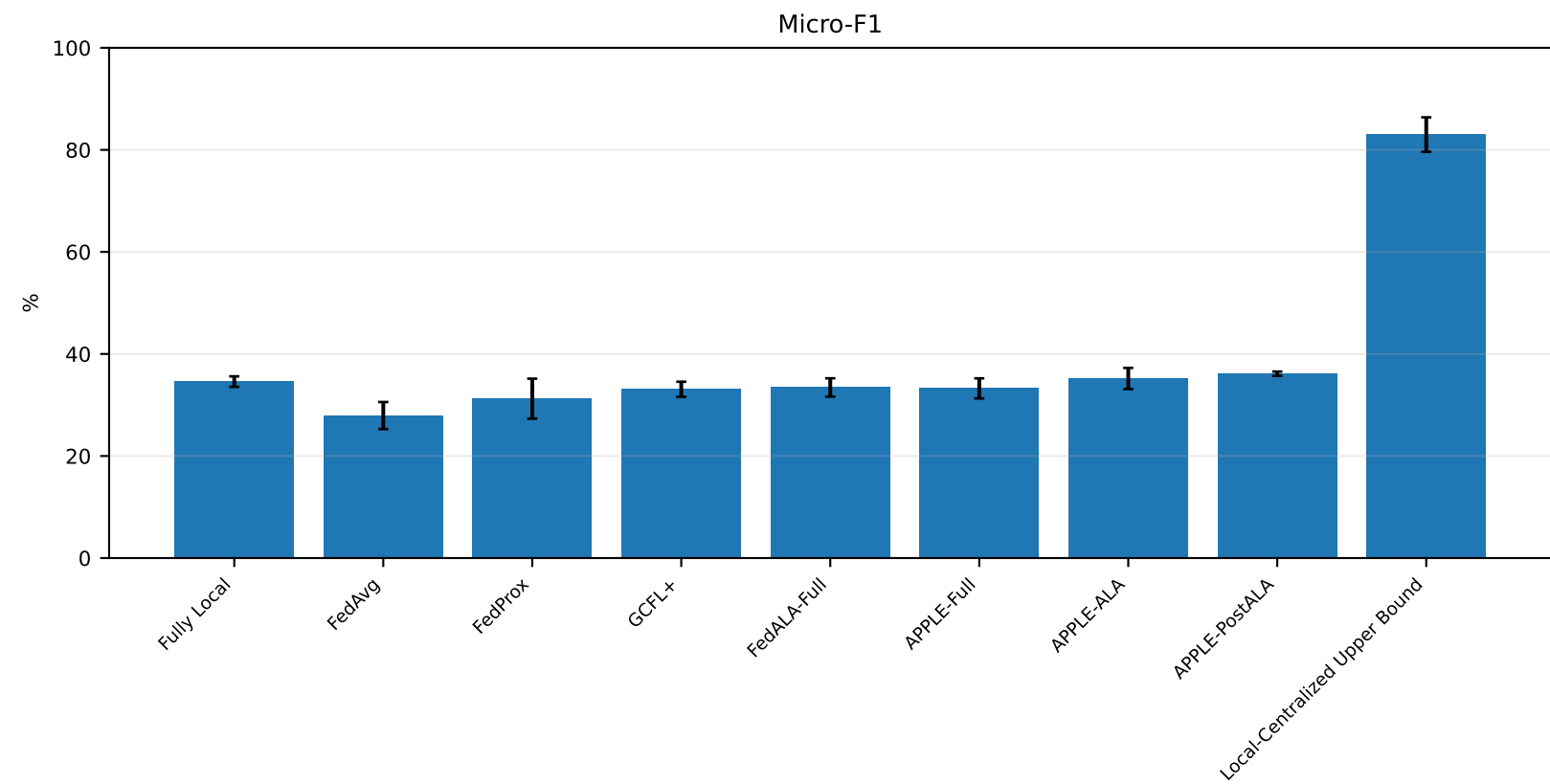
Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pr_auc | protocol=OO | overview bars
O+O: oracle/full validation selection → oracle/full test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pr_auc | protocol=VV
V+V: visible validation selection → visible test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	34.6% ± 1.0%	41.2% ± 2.0%	51.7% ± 3.5%	47.3% ± 1.6%	61.0% ± 6.7%	55.6% ± 3.9%	35.5% ± 2.8%	33.6% ± 4.2%	20.5% ± 1.1%	—	3
FedAvg	27.9% ± 2.7%	30.5% ± 3.2%	53.2% ± 1.9%	50.6% ± 0.7%	41.8% ± 9.0%	34.9% ± 3.2%	28.5% ± 3.8%	27.1% ± 3.7%	20.5% ± 2.6%	67.7 ± 15.0	3
FedProx	31.2% ± 3.9%	35.3% ± 4.2%	56.4% ± 0.7%	53.4% ± 0.6%	49.3% ± 4.0%	45.3% ± 9.0%	30.3% ± 0.8%	29.7% ± 4.8%	21.9% ± 4.0%	75.7 ± 3.8	3
GCFL+	33.1% ± 1.5%	36.7% ± 2.5%	60.6% ± 9.2%	57.7% ± 7.8%	46.5% ± 10.4%	47.5% ± 2.5%	35.2% ± 2.2%	31.3% ± 2.9%	22.9% ± 0.6%	69.7 ± 4.0	3
FedALA-Full	33.4% ± 1.8%	38.3% ± 2.6%	55.4% ± 13.1%	53.4% ± 11.3%	58.2% ± 1.5%	46.9% ± 7.9%	34.0% ± 3.1%	30.5% ± 2.2%	22.0% ± 1.2%	74.3 ± 4.0	3
APPLE-Full	33.3% ± 2.0%	38.0% ± 3.5%	48.6% ± 4.1%	48.9% ± 2.9%	52.6% ± 17.6%	50.9% ± 1.0%	29.8% ± 3.8%	33.9% ± 0.7%	22.8% ± 2.0%	68.3 ± 11.7	3
APPLE-ALA	35.2% ± 2.1%	42.1% ± 3.8%	58.2% ± 9.5%	55.6% ± 7.5%	68.5% ± 10.1%	51.7% ± 7.9%	37.5% ± 1.5%	31.1% ± 0.8%	21.6% ± 0.3%	71.3 ± 11.7	3
APPLE-PostALA	36.1% ± 0.4%	44.0% ± 2.9%	59.7% ± 8.9%	58.0% ± 6.6%	68.9% ± 11.3%	57.2% ± 4.6%	38.8% ± 2.9%	33.0% ± 0.9%	21.8% ± 1.6%	65.3 ± 13.7	3
Local-Centralized Upper Bound	83.0% ± 3.4%	85.7% ± 3.5%	95.0% ± 1.0%	93.7% ± 0.7%	97.3% ± 1.3%	96.5% ± 1.3%	86.8% ± 8.5%	79.8% ± 8.2%	67.9% ± 2.0%	—	3

Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pr_auc | protocol=VV | overview bars
V+V: visible validation selection → visible test



Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity (q=0.5) | selector=macro_pr_auc | protocol=VO
V+O: visible validation selection → oracle/full test

Method	Micro-F1	Macro Pos-F1	Micro PR-AUC	Macro PR-AUC	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1	Sel. round	n
Fully Local	71.3% ± 0.9%	74.5% ± 1.1%	82.4% ± 0.6%	80.1% ± 0.3%	89.6% ± 2.8%	86.4% ± 1.5%	73.6% ± 1.8%	70.8% ± 3.8%	52.0% ± 0.9%	—	3
FedAvg	66.0% ± 3.1%	67.8% ± 3.1%	83.1% ± 1.0%	82.3% ± 1.1%	77.9% ± 5.5%	73.2% ± 2.6%	66.7% ± 4.3%	65.0% ± 4.5%	56.1% ± 3.9%	67.7 ± 15.0	3
FedProx	69.6% ± 3.4%	71.8% ± 3.0%	85.3% ± 0.7%	84.3% ± 1.2%	83.6% ± 0.9%	80.7% ± 4.9%	69.1% ± 0.7%	67.7% ± 4.5%	58.0% ± 5.4%	75.7 ± 3.8	3
GCFL+	71.2% ± 1.7%	72.9% ± 1.9%	86.2% ± 4.1%	85.8% ± 3.5%	80.0% ± 5.9%	81.8% ± 1.5%	73.1% ± 2.6%	69.7% ± 2.9%	59.8% ± 0.9%	69.7 ± 4.0	3
FedALA-Full	70.5% ± 2.1%	72.8% ± 2.2%	83.3% ± 5.2%	82.9% ± 4.7%	86.1% ± 2.2%	80.4% ± 4.7%	71.0% ± 2.9%	68.3% ± 2.7%	58.1% ± 1.8%	74.3 ± 4.0	3
APPLE-Full	70.6% ± 2.2%	72.6% ± 2.6%	79.8% ± 2.8%	80.4% ± 2.4%	82.3% ± 9.4%	83.2% ± 0.7%	67.4% ± 4.1%	71.3% ± 1.4%	59.0% ± 2.5%	68.3 ± 11.7	3
APPLE-ALA	72.1% ± 1.7%	75.0% ± 2.1%	85.1% ± 3.8%	83.5% ± 2.4%	91.1% ± 4.7%	83.4% ± 4.9%	75.1% ± 1.3%	68.4% ± 1.0%	57.2% ± 0.6%	71.3 ± 11.7	3
APPLE-PostALA	73.5% ± 0.3%	76.5% ± 0.9%	86.1% ± 2.6%	85.1% ± 2.0%	90.3% ± 5.8%	87.6% ± 1.8%	75.9% ± 2.4%	70.8% ± 1.3%	57.9% ± 3.1%	65.3 ± 13.7	3
Local-Centralized Upper Bound	83.3% ± 2.7%	86.0% ± 3.0%	95.8% ± 0.9%	94.3% ± 0.5%	98.4% ± 1.8%	96.3% ± 1.5%	86.2% ± 7.8%	80.3% ± 7.1%	68.6% ± 3.2%	—	3

Main baselines + APPLE/FedALA/PostALA | Mid heterogeneity ($q=0.5$) | selector=macro_pr_auc | protocol=VO | overview bars
V+O: visible validation selection → oracle/full test

